

DASC7606

Deep Learning

4: ImageNet

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2025/26

Convolutional Neural Networks (CNNs)

1. What is Computer Vision?
2. Milestones in Computer Vision
3. What is Convolutional Neural Networks (CNNs)?
 - a. Convolutional layer – Filter
 - b. Pooling
 - c. Dense layer
4. Image Classification
5. AlexNet 2012 and ZFNet 2013
6. VGGNet 2014
7. GoogLeNet 2014
8. ResNet 2015
9. ReduNet 2022

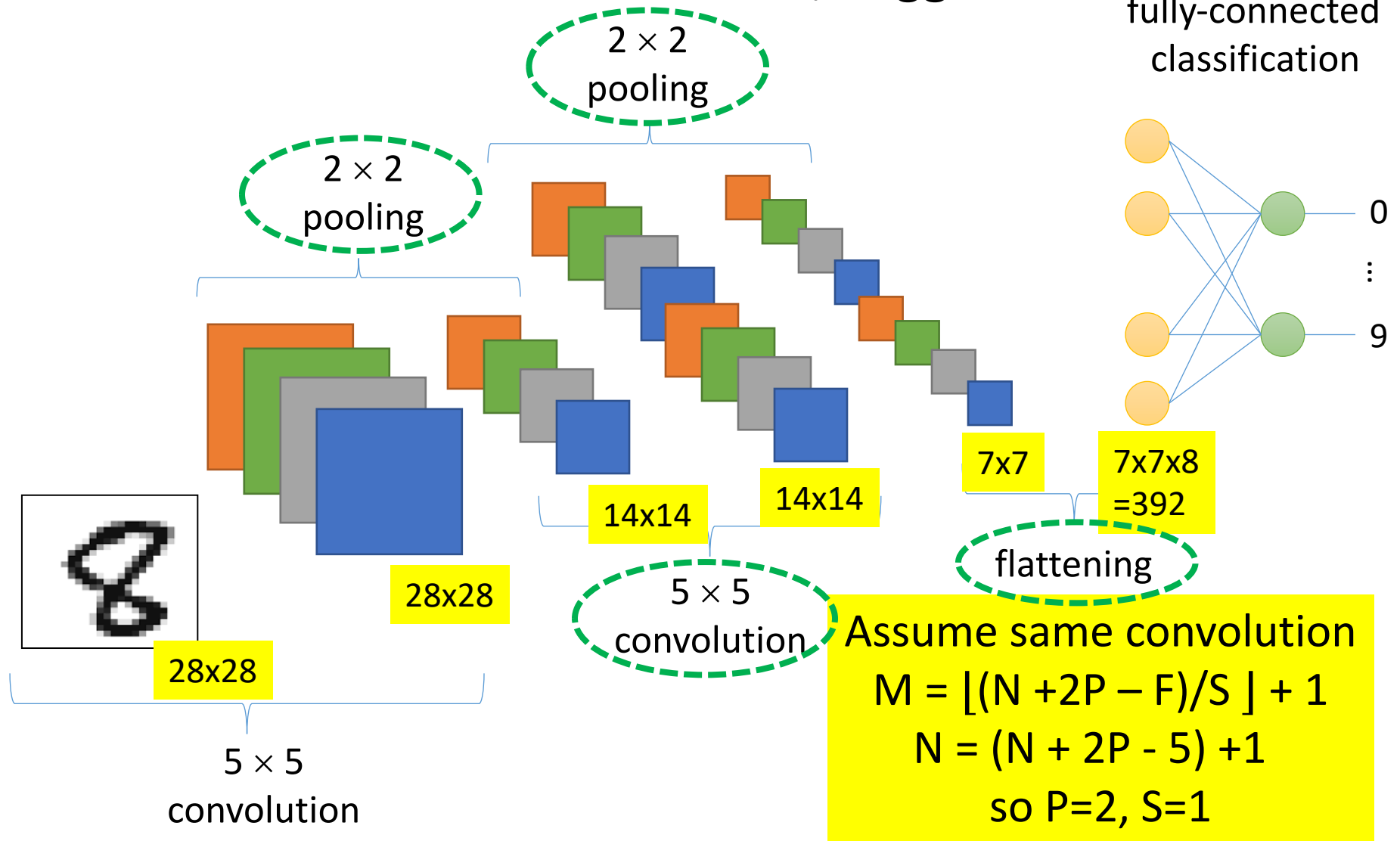
References

- MIT online course 6.S191 “Introduction to Deep Learning”
- Stanford CS230 “Deep Learning”
- Chan, Yu, You, Qi, Wright, Ma, ReduNet: A White-box Deep Network from the Principle of Maximizing Rate Reduction, Journal of Machine Learning Research 23 (2022) 1-103

Exercise

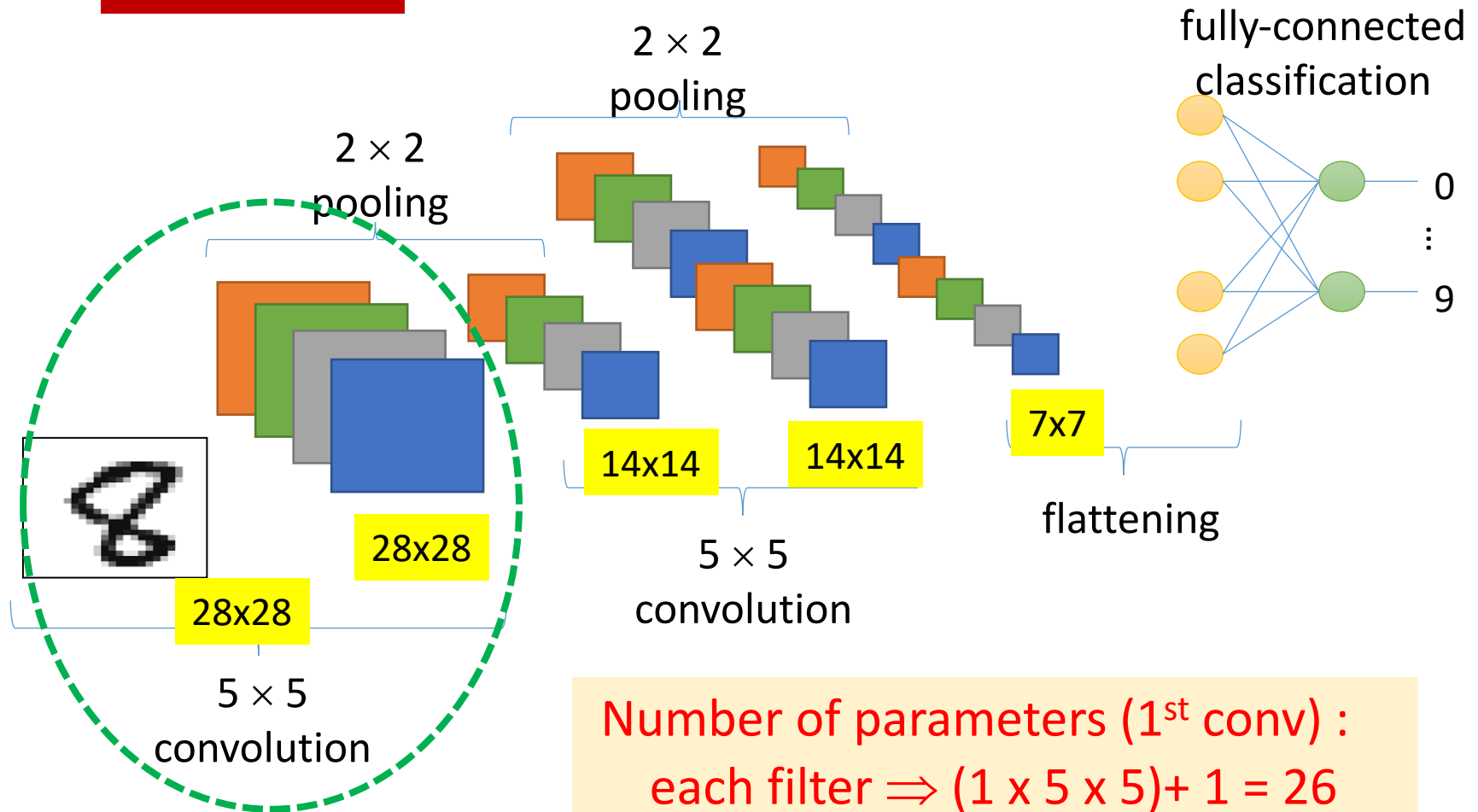
Size of the Network

The best known **MNIST** classifier, Kaggle



Exercise

No of Parameters and Computation



Total Number
of Neurons

Number of parameters (1st conv) :

each filter $\Rightarrow (1 \times 5 \times 5) + 1 = 26$

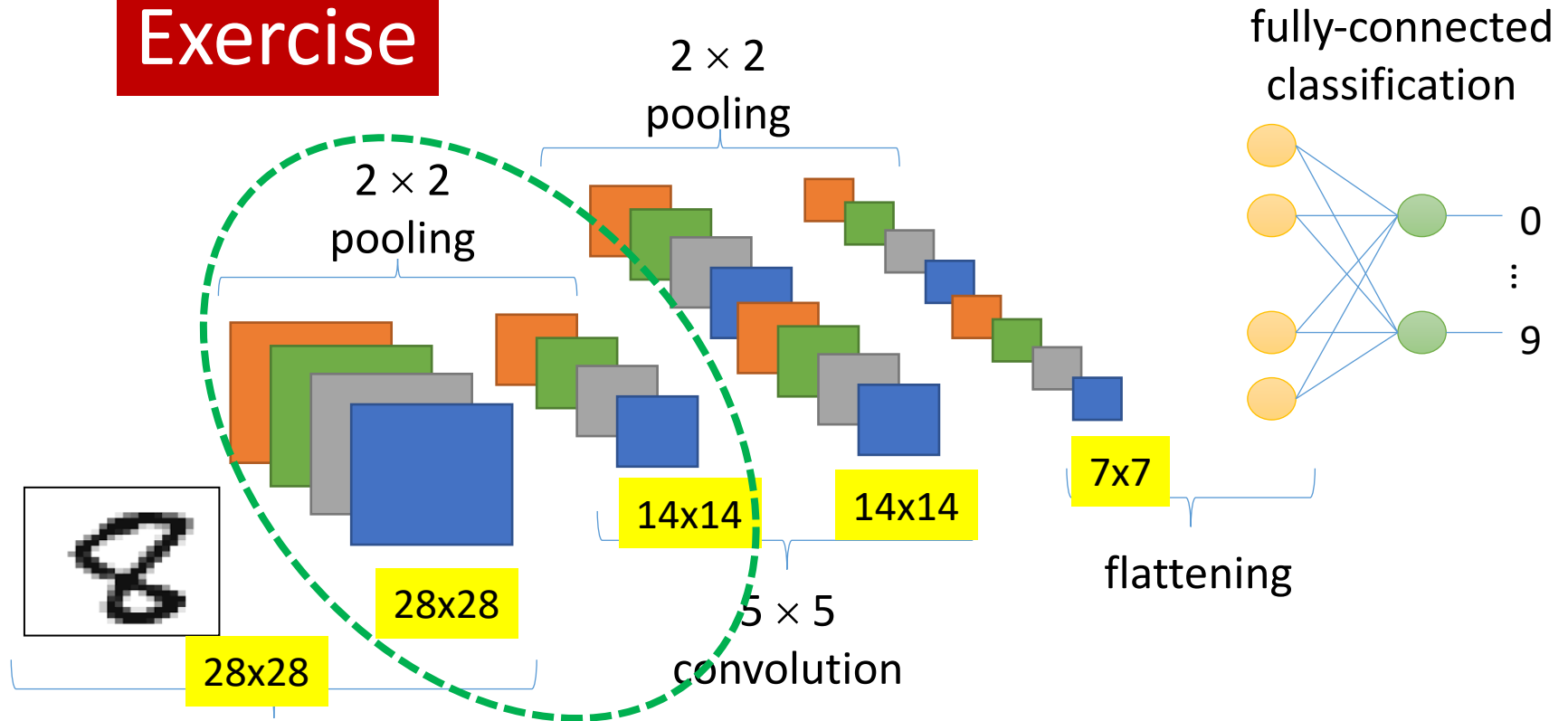
4 channels $\Rightarrow 4 \times 26 = 104$

Computation (multiplications)

$(5 \times 5) \times (4 \times 28 \times 28) = 78400$

LRF = 5x5

Exercise



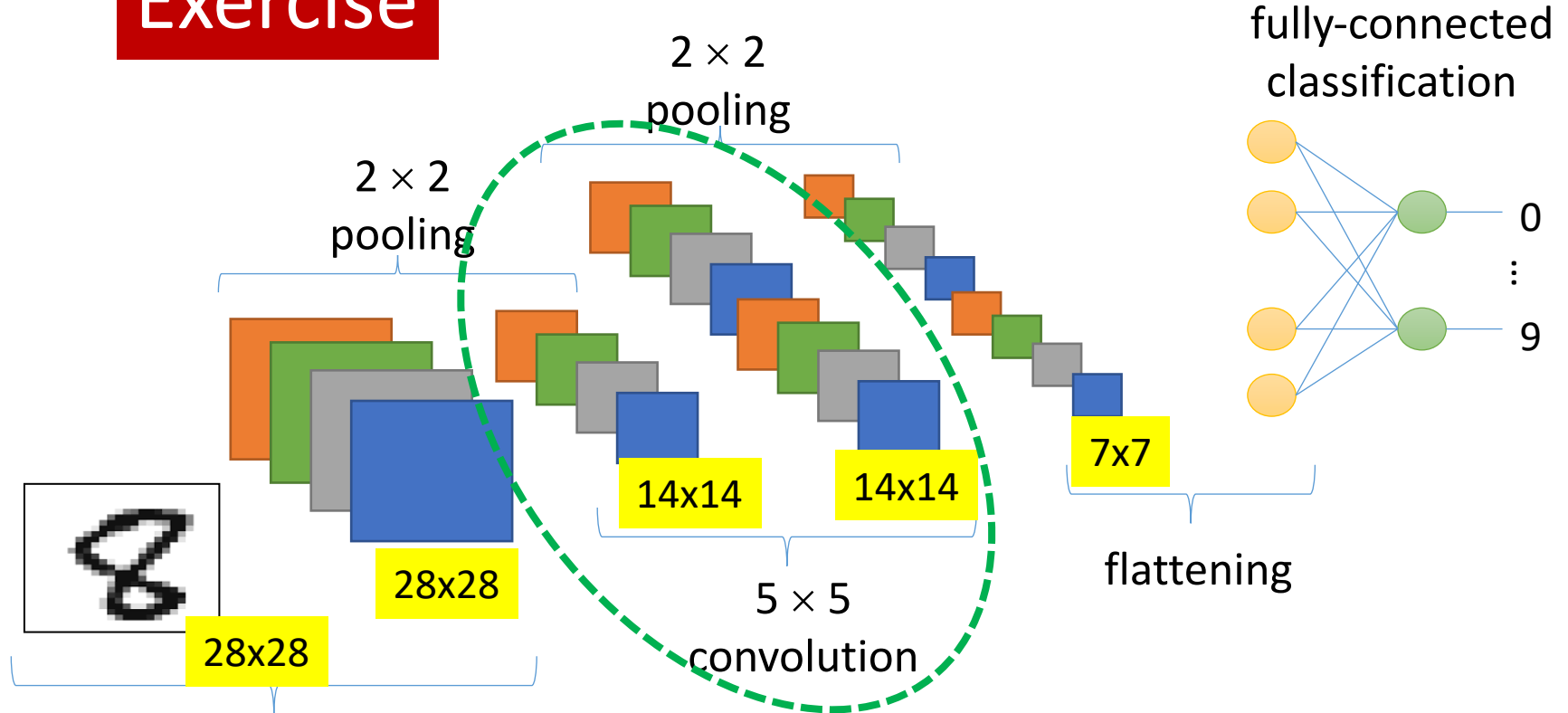
Pooling (Max)

Number of parameters : none

Computation (multiplications) : none

LRF = 6x6

Exercise



Number of parameters (2nd conv)

each filter $\Rightarrow 4 \times 5 \times 5 + 1 = 101$

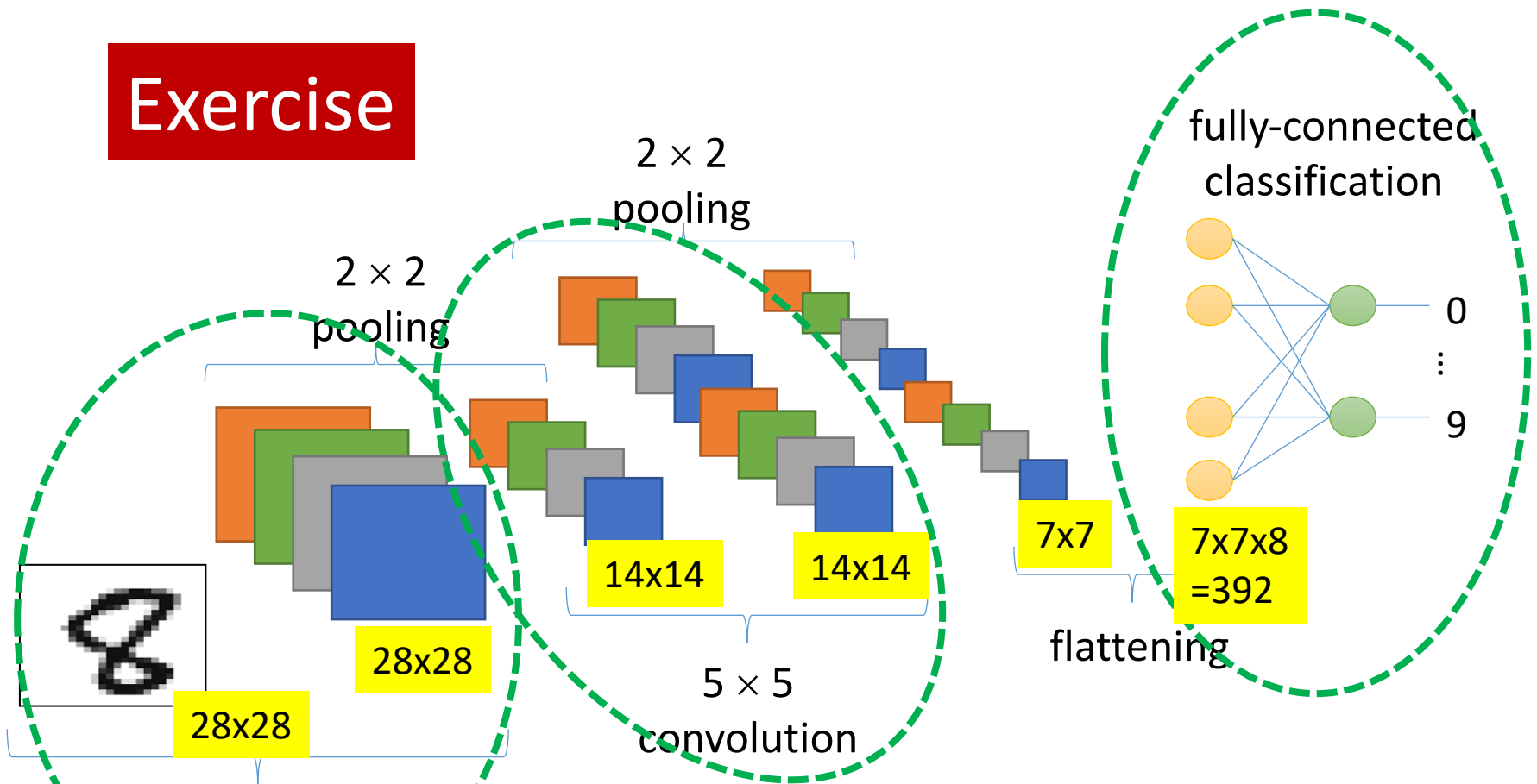
8 filters $\Rightarrow 8 \times (4 \times 5 \times 5) + 8 = 808$

Computation

$(4 \times 5 \times 5) \times (8 \times 14 \times 14) = 156,800$

LRF = 14x14

Exercise



Number of parameters

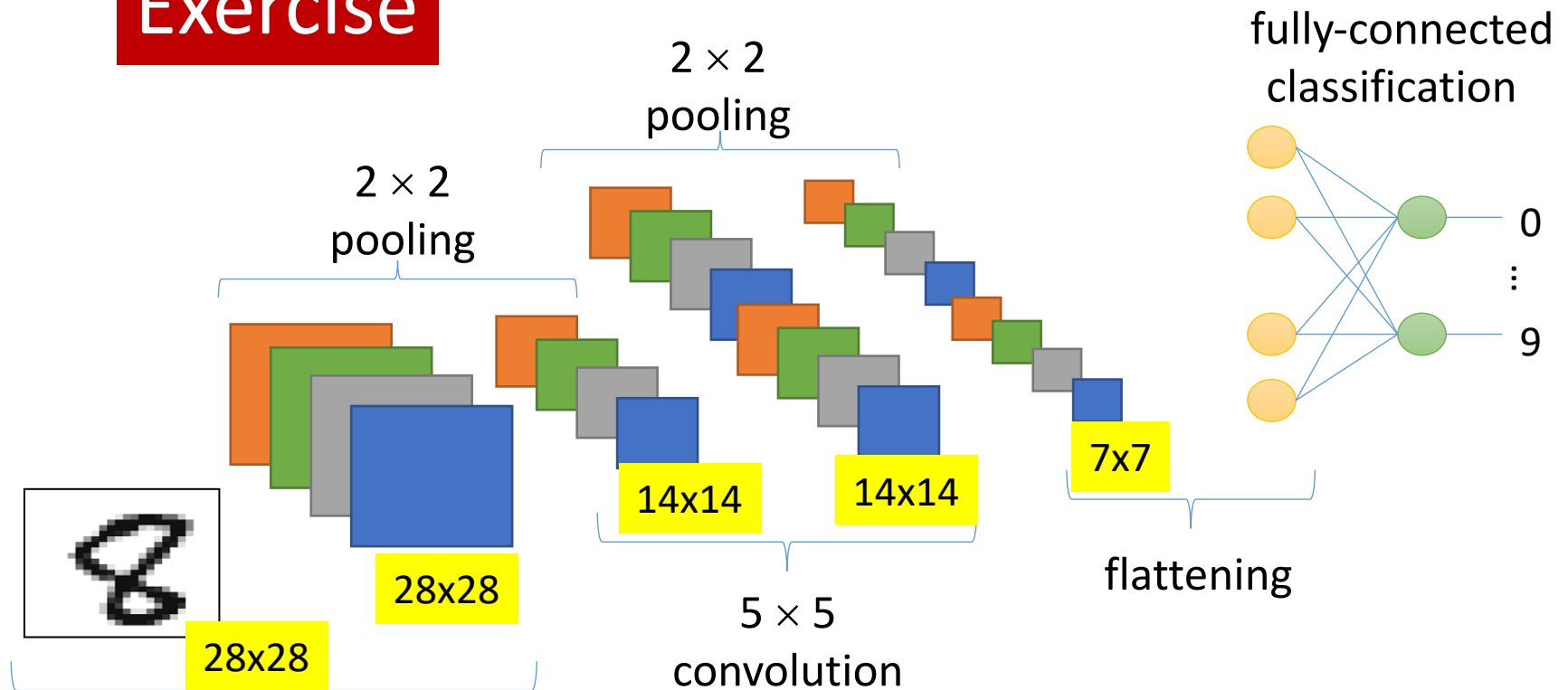
1st conv : 4 filters $\Rightarrow 4 \times (1 \times 5 \times 5 + 1) = 104$

2nd conv: 8 filters $\Rightarrow 8 \times (4 \times 5 \times 5 + 1) = 808$

Dense : $\Rightarrow (8 \times 7 \times 7 + 1) \times 10 = 3,930$

TOTAL : $\Rightarrow 4,842$

Exercise



Computation (multiplications) / Connection

1st conv : 4 filters $\Rightarrow 5 \times 5 \times 4 \times 28 \times 28 = 78,400$

2nd conv: 8 filters $\Rightarrow 4 \times 5 \times 5 \times 8 \times 14 \times 14 = 156,800$

Dense : $\Rightarrow 8 \times 7 \times 7 \times 10 = 3,920$

TOTAL $\Rightarrow 239,120$

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4. A Simple Image Classifier for CIFAR-10 (2013)

CIFAR-10(0): tiny images

- 60000 32x32 color images in 10 classes
(plane, car, bird, cat, deer, dog, frog, horse, ship, truck)

K-Nearest Neighbors Image Diff Classifier

- Distance measure: Pixel-wise value difference

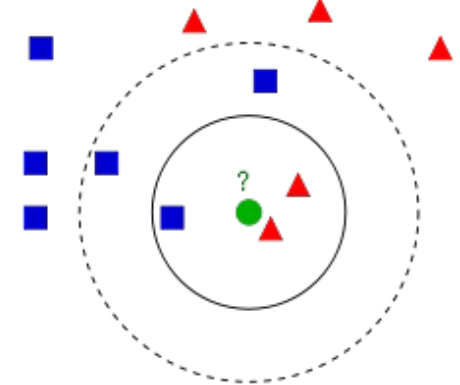
test image				training image				pixel-wise absolute value differences			
56	32	10	18	10	20	24	17	46	12	14	1
90	23	128	133	8	10	89	100	82	13	39	33
24	26	178	200	12	16	178	170	12	10	0	30
2	0	255	220	4	32	233	112	2	32	22	108

-

=

→ 456

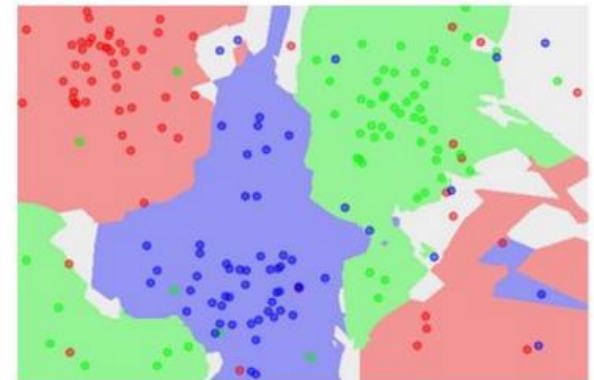
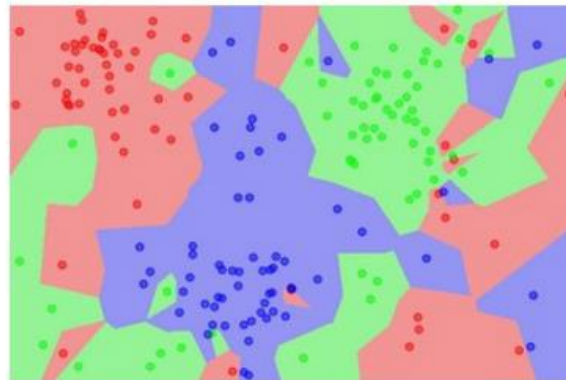
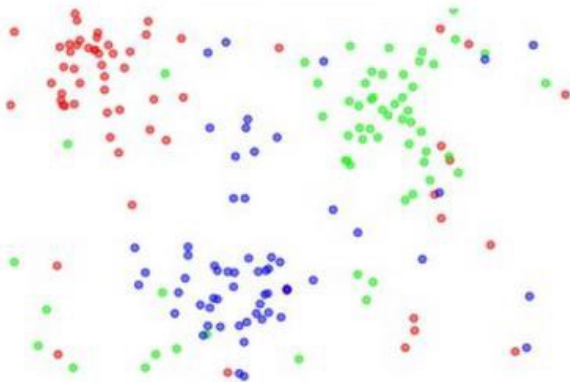
Choosing K = 1,3,5?



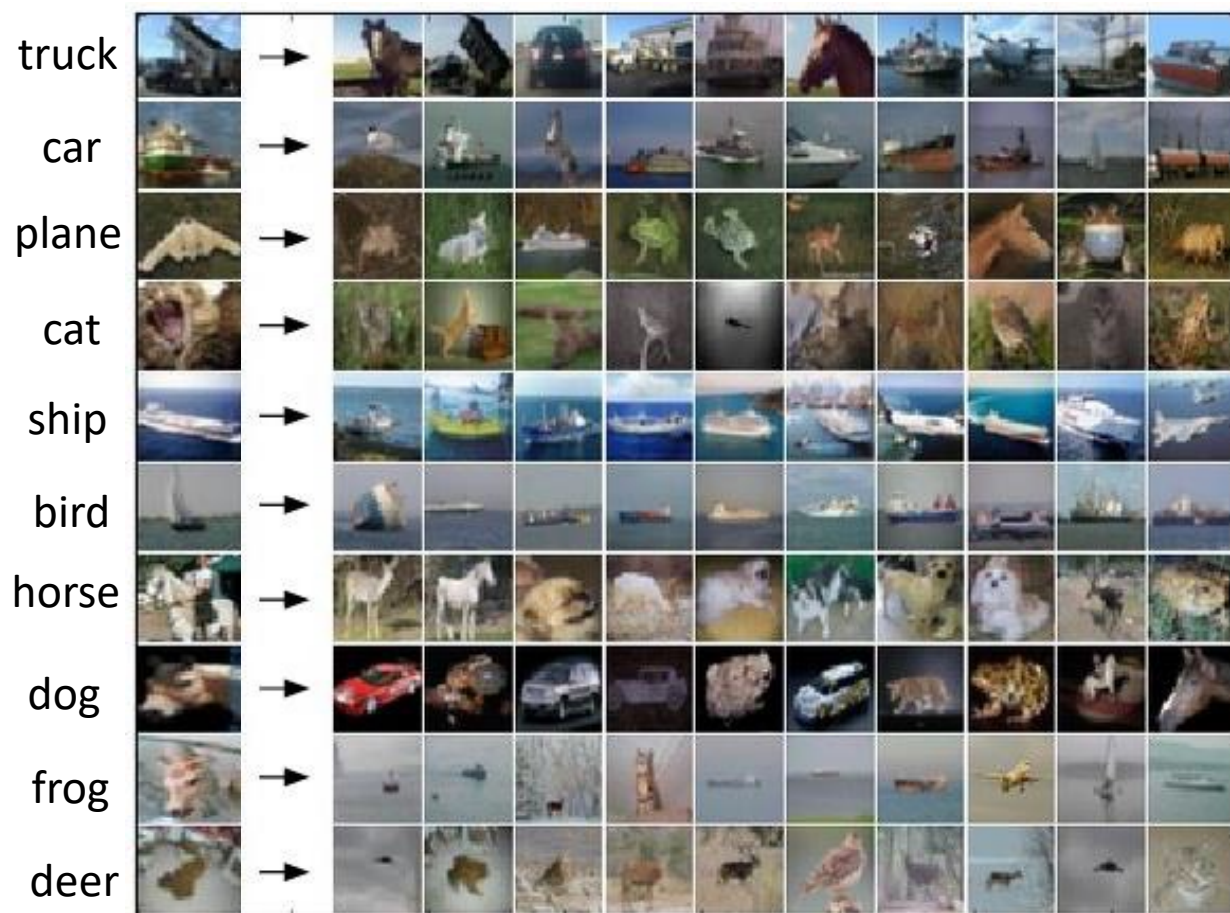
the data

NN classifier

5-NN classifier



A Simple Image Classifier for CIFAR-10 (2013)



Accuracy

Random: 10%

Image difference

(L1): 38.6%

(L2): 35.4%

Human: 94%

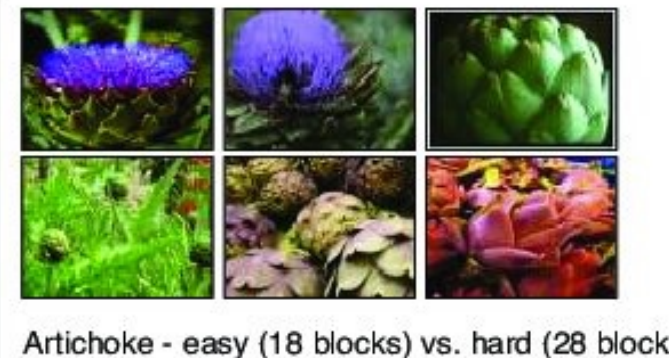
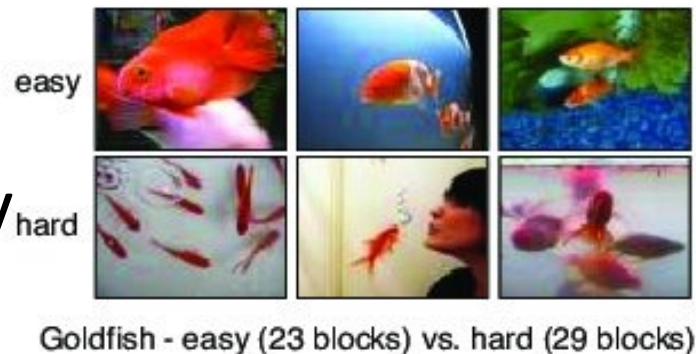
CNN: 95%

ImageNet (2010)

- ~15M images collected from web
took 3 yrs to organize and label
- Labelled into ~ 22K categories:
Animals (bird, fish, mammal, ...),
Plants, Food, Person, Sports (etc.).
- www.image-net.org



“Are we ready
to recognize
all objects?”

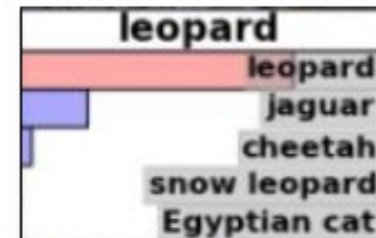


ImageNet Large-Scale Visual Recognition Challenge (ILSVRC)

- **annual competition** of image **classification**:
- 1.2M images in 1K categories - **5 guesses** on label



Classification



Top-5 error rate



Bear
Whale
Bird
Car
Train

Yes!

Dolphin
Bear
Bird
Car
Train

No!

Classifications not easy even for humans



EntleBucher



Appenzeller



mite



container ship



motor scooter



leopard

	mite
	black widow
	cockroach
	tick
	starfish

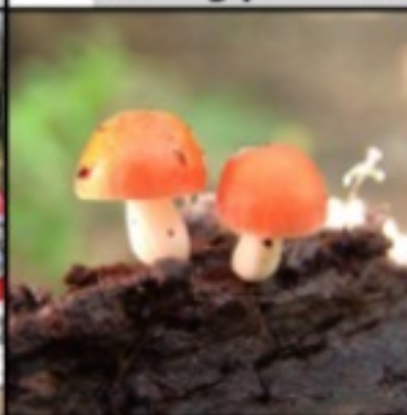
	container ship
	lifeboat
	amphibian
	fireboat
	drilling platform

	motor scooter
	go-kart
	moped
	bumper car
	golfcart

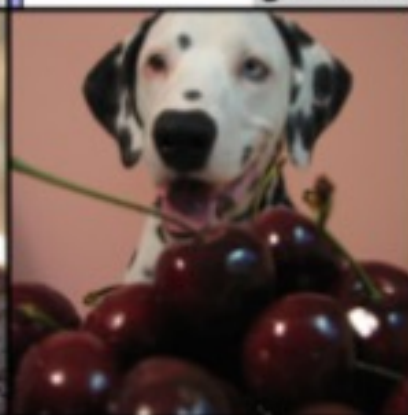
	leopard
	jaguar
	cheetah
	snow leopard
	Egyptian cat



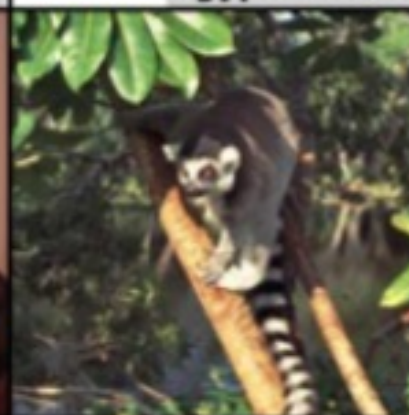
grille



mushroom



cherry



Madagascar cat

	convertible
	grille
	pickup
	beach wagon
	fire engine

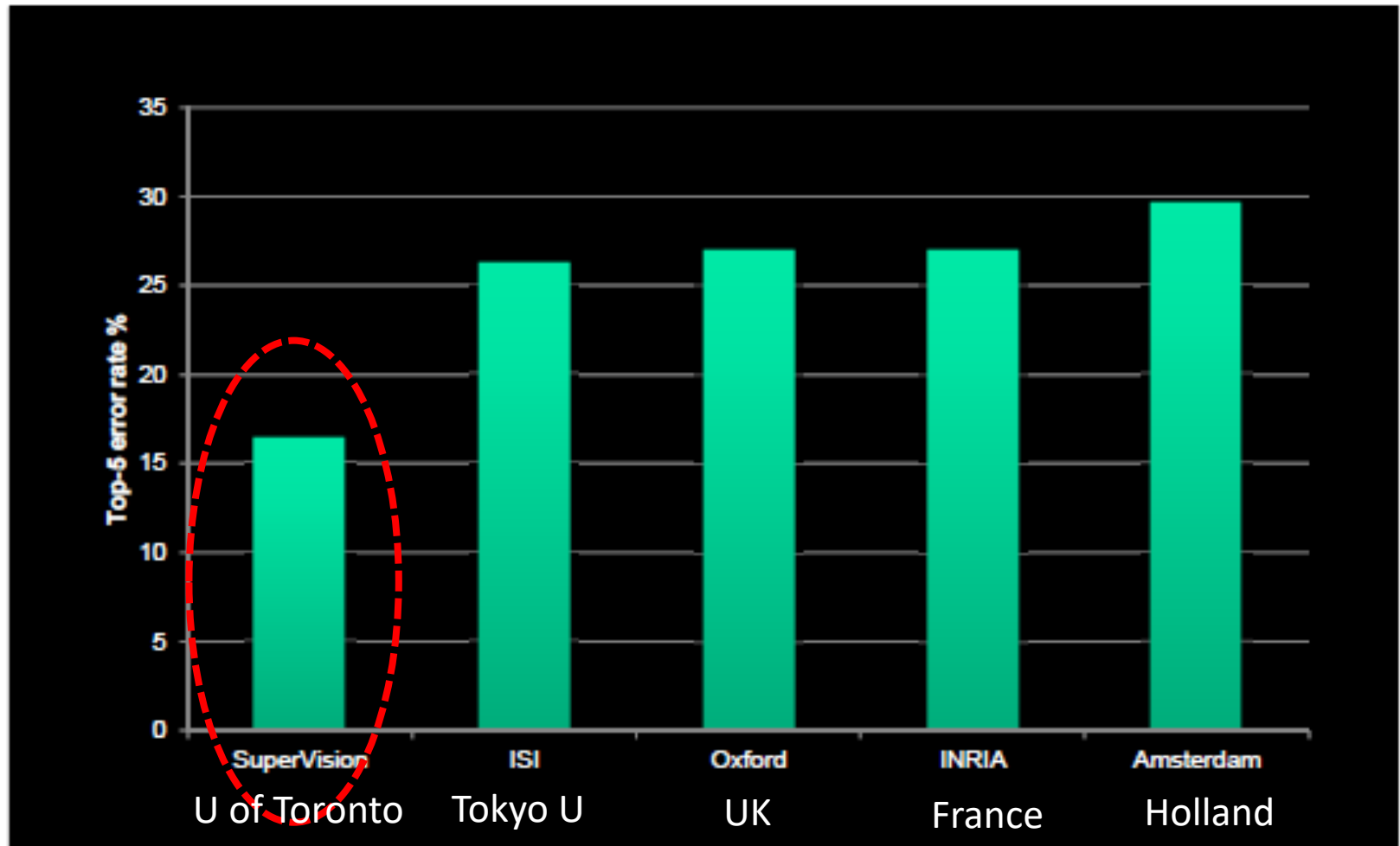
	agaric
	mushroom
	jelly fungus
	gill fungus
	dead-man's-fingers

	dalmatian
	grape
	elderberry
	ffordshire bullterrier
	currant

	squirrel monkey
	spider monkey
	titi
	indri
	howler monkey

AlexNet/SuperVision (2012) –

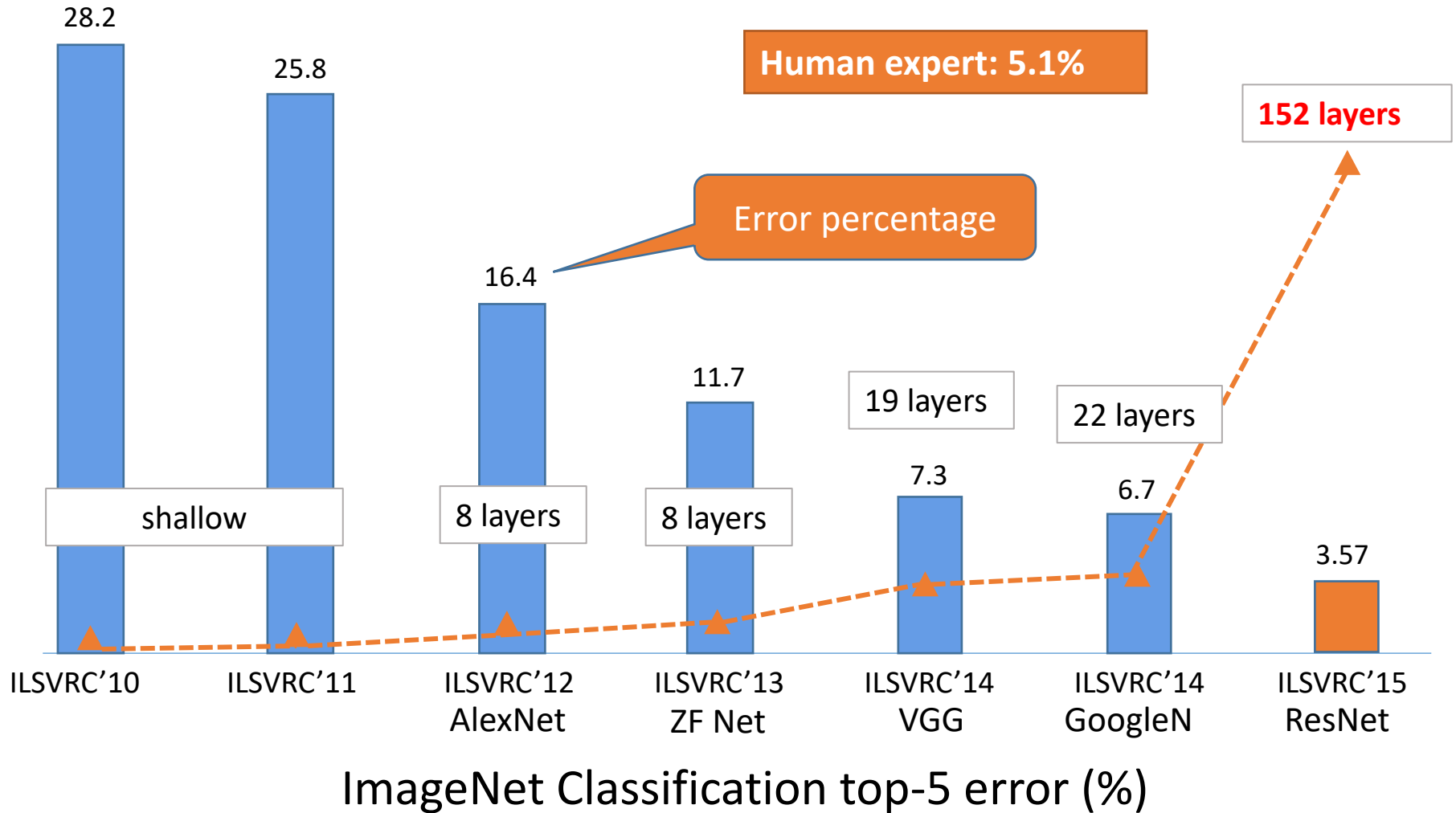
“ImageNet Classification with Deep Convolutional Networks”
similar to Yann LeCun’s LeNet-5 (1998) → 16.4%



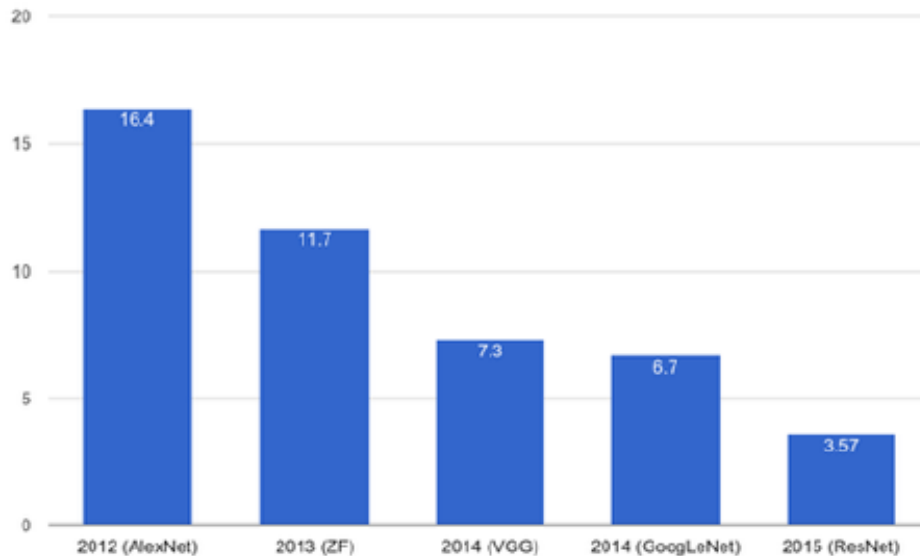
Alex Krizhevsky, I Sutskever, GE Hinton, ImageNet Classification with Deep Convolutional Neural Networks

Geoffrey Hinton, 2024 Nobel prize winner

Revolution of Depth



ImageNet Classification Error (Top 5)



- **AlexNet (2012): First CNN (15.4%)**
 - 8 layers
 - 61 million parameters
- **ZFNet (2013): 15.4% to 11.2%**
 - 8 layers
 - More filters. Denser stride.
- **VGGNet (2014): 11.2% to 7.3%**
 - Beautifully uniform:
3x3 conv, stride 1, pad 1, 2x2 max pool
 - 16 layers
 - 138 million parameters
- **GoogLeNet (2014): 11.2% to 6.7%**
 - Inception modules
 - 22 layers
 - 5 million parameters
(throw away fully connected layers)
- **ResNet (2015): 6.7% to 3.57%**
 - More layers = better performance
 - 152 layers
- **CUIImage (2016): 3.57% to 2.99%**
 - Ensemble of 6 models

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AlexNet (2012)

- 7 hidden layers
- 650K neurons
- 60M parameters
- 630M connections
- Overlapping Max Pooling (size 3×3 , stride 2)
- Rectified Linear Units (ReLU)
 - faster than tanh and sigmoid activation functions
- Dropout trick (on fully-connected), batched SGD
- Data augmentation introduces variations
e.g., randomly extracted 227×227 patches
- Two GPUs, training in 5-6 days

Source: Alex Krizhevsky, et. al., ImageNet Classification with Deep Convolutional Neural Networks, CACM, 2017
[h-p://image-net.org/challenges/LSVRC/2012/supervision.pdf](http://image-net.org/challenges/LSVRC/2012/supervision.pdf)

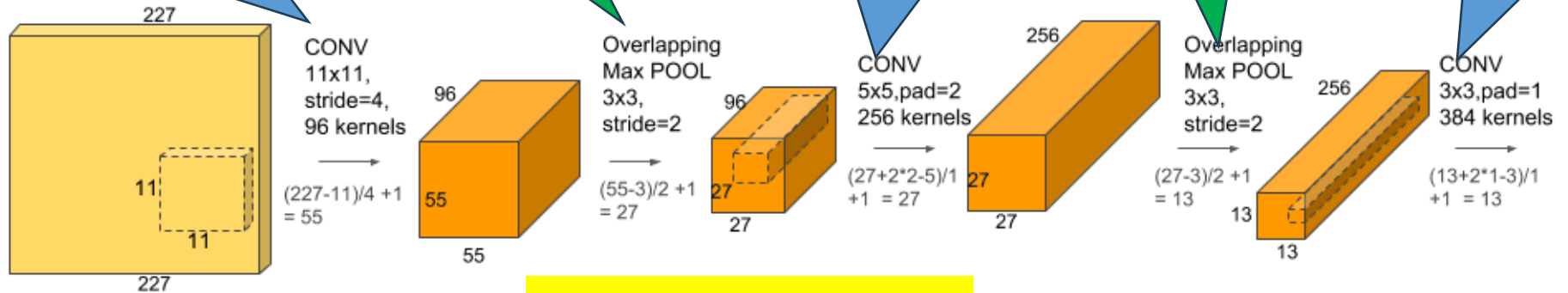
Convolution
96 filters
11 x 11, S=4
 $(227-11)/4 + 1$
= 55

Pooling
3x3 S=2
 $(55-3)/2 + 1$
= 27

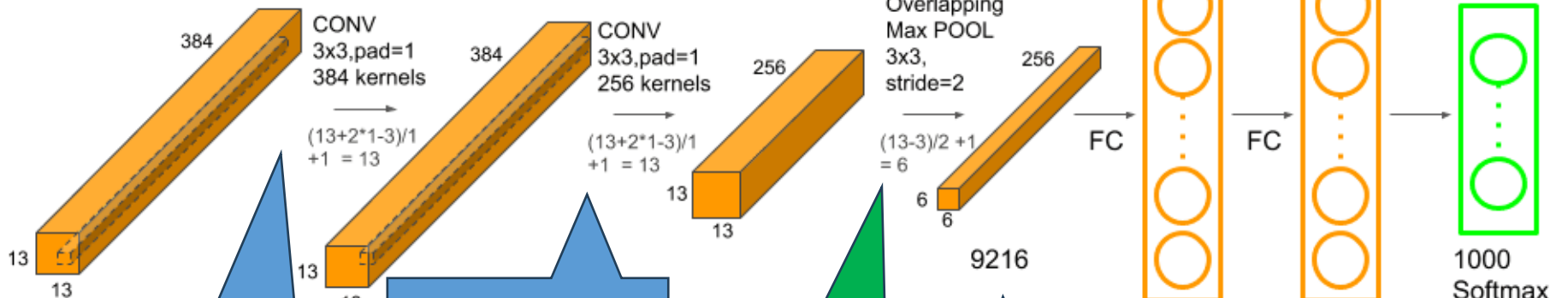
Convolution
256 filters
5x5, pad 2
 $(27+4-5)/1 + 1$
= 27

Pooling
3x3 S=2
 $(27-3)/2 + 1$
= 13

Convolution
384 filters
3x3, pad 1
 $(13+2-3)/1 + 1$
= 13



$$M = \lfloor (N + 2P - F) / S \rfloor + 1$$



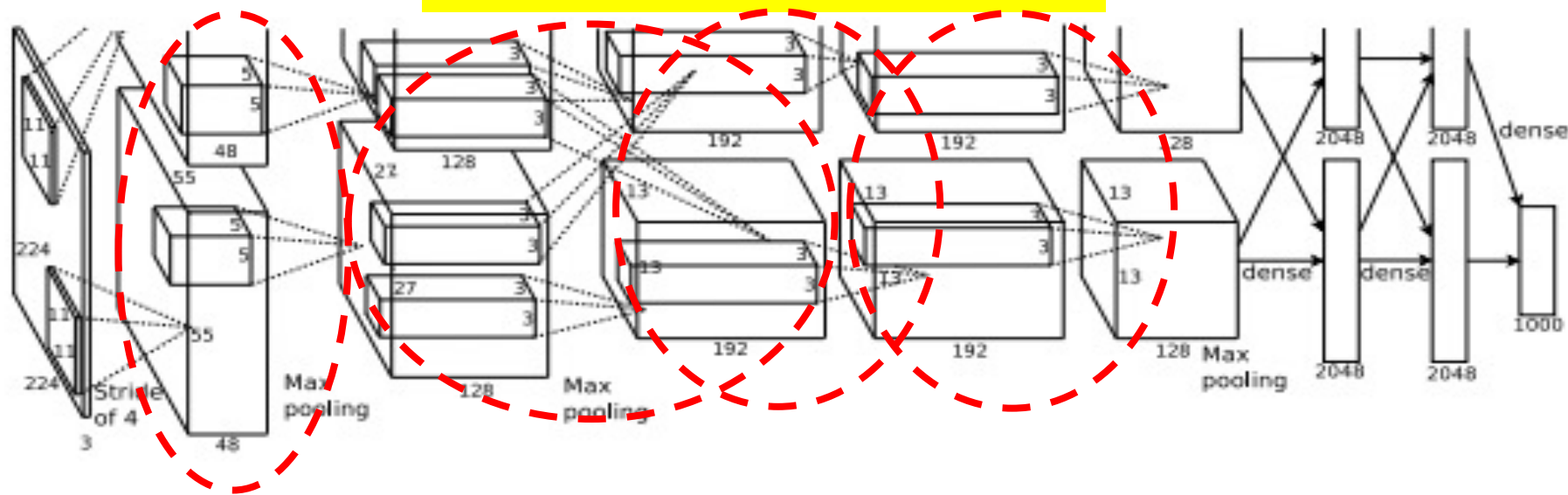
Convolution
384 filters
3x3, pad 1
 $(13+2-3)/1 + 1 = 13$

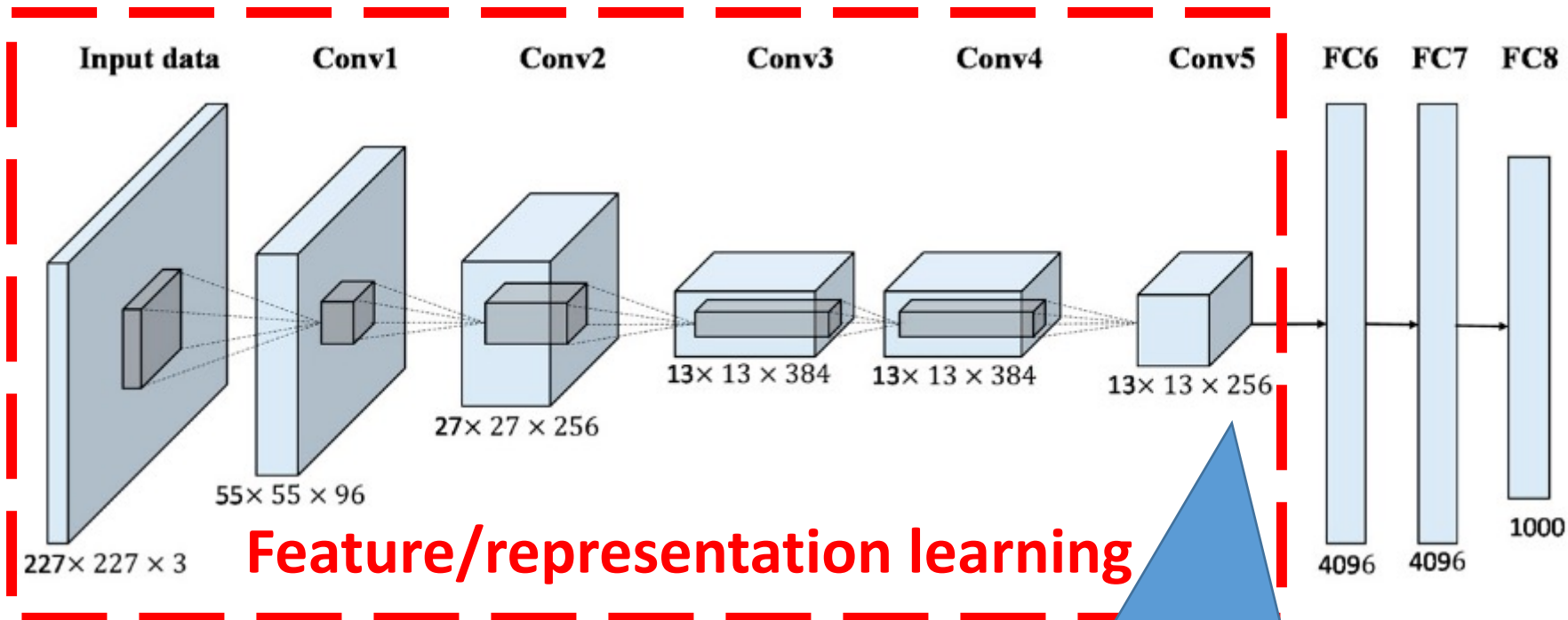
Convolution
256 filters
3x3, pad 1
 $(13+2-3)/1 + 1 = 13$

Pooling
3x3 S=2
 $(13-3)/2 + 1$
= 6

Flattening
256 x 6 x 6
= 9216

1000
Softmax





Embedding:

Translate large sparse vectors into low-dimensional space (vectors) that preserve semantic relationship.

Image

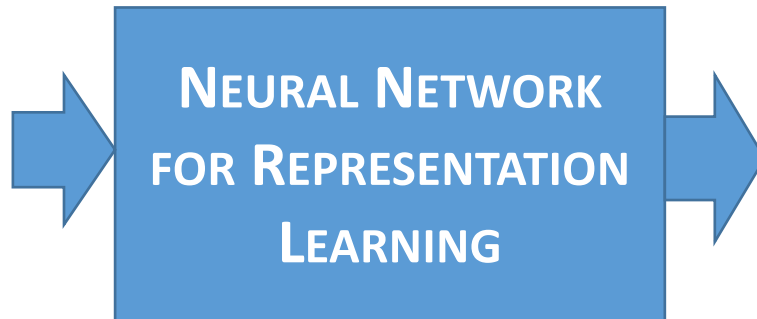
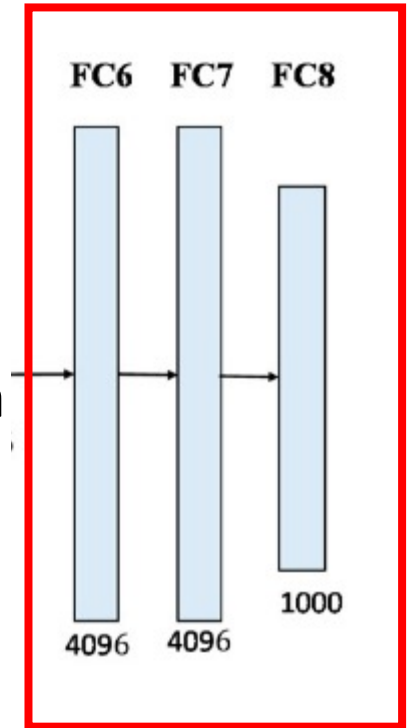


Image Representation



This part might look different for:

- Different image classification **domains**
- Image captioning with **recurrent neural networks**
- Image object localization with **bounding box**
- Image segmentation with **fully convolutional networks**
- Image segmentation with **deconvolution layers**

ZFNet (2013)

Zeiler and Fergus, Visualizing and Understanding Convolutional Networks, 2014

In 2013 ZFNet achieved 11.7% error rate.

ZFNet basically focused on tuning the hyperparameters of AlexNet.

- by unpool and deconv each layer for visualization
- aliasing and missing mid frequencies problems because low sampling rate.

Same architecture but:

- CONV1: change from 11x11 stride 4, to 7x7 stride 2
- CONV3,4,5: 384, 384, 256 filters to 512, 1024, 512.

The main contribution is stated in the title

“Visualizing and Understanding Convolutional Networks” which will be discussed later.

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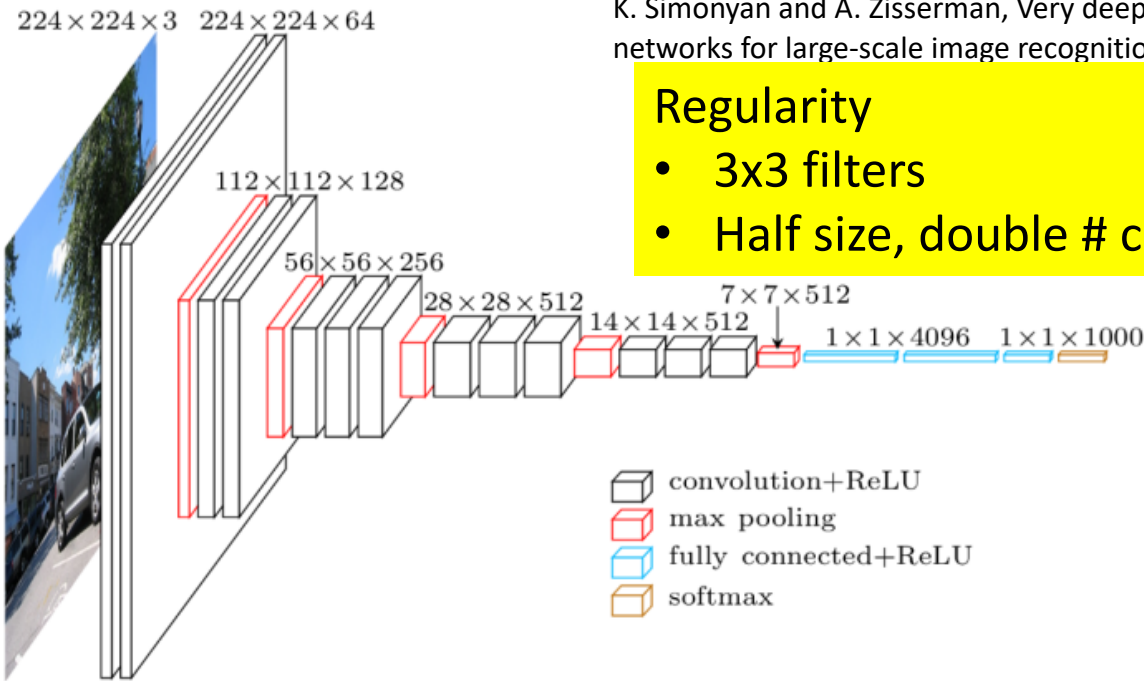
- MIT online course 6.S191 “Introduction to Deep Learning”
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VGGNet 2014 (Oxford Visual Geometry Group)

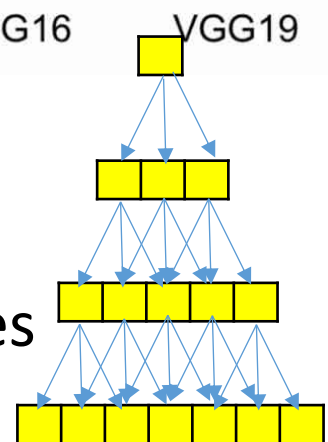
K. Simonyan and A. Zisserman, Very deep convolutional networks for large-scale image recognition, 2015

Regularity

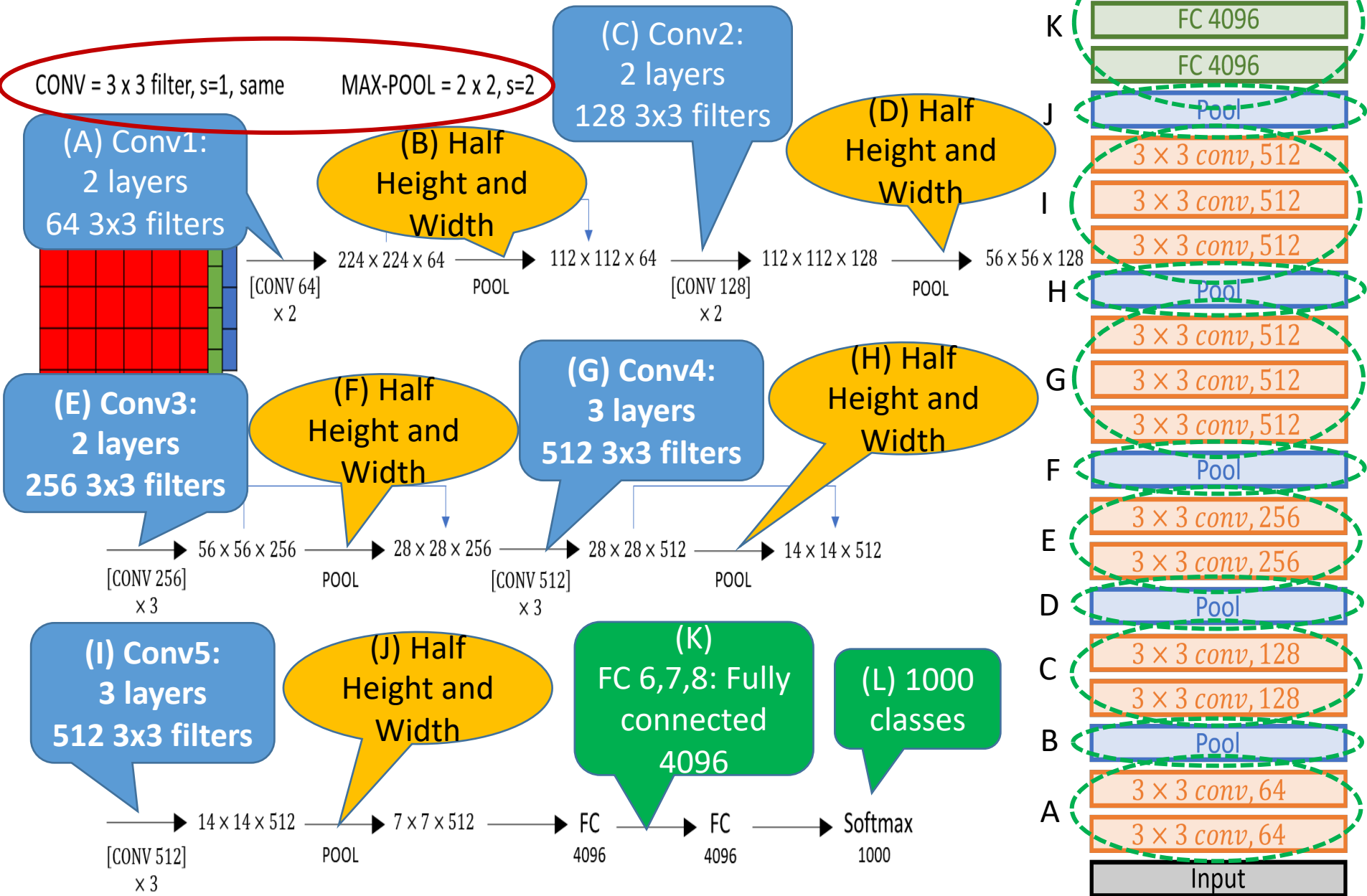
- 3x3 filters
- Half size, double # channels



- 2nd classification, 1st localization
- Three 3x3 conv layer (stride 1) has the same effective local receptive field (LRF) as one 7x7 conv layer but has less parameters, $3 \times (3 \times 3 \times c) < (7 \times 7 \times c)$
- More layers (VGG16/19) means more non-linearities
- About 138M parameters (most in FC layers)

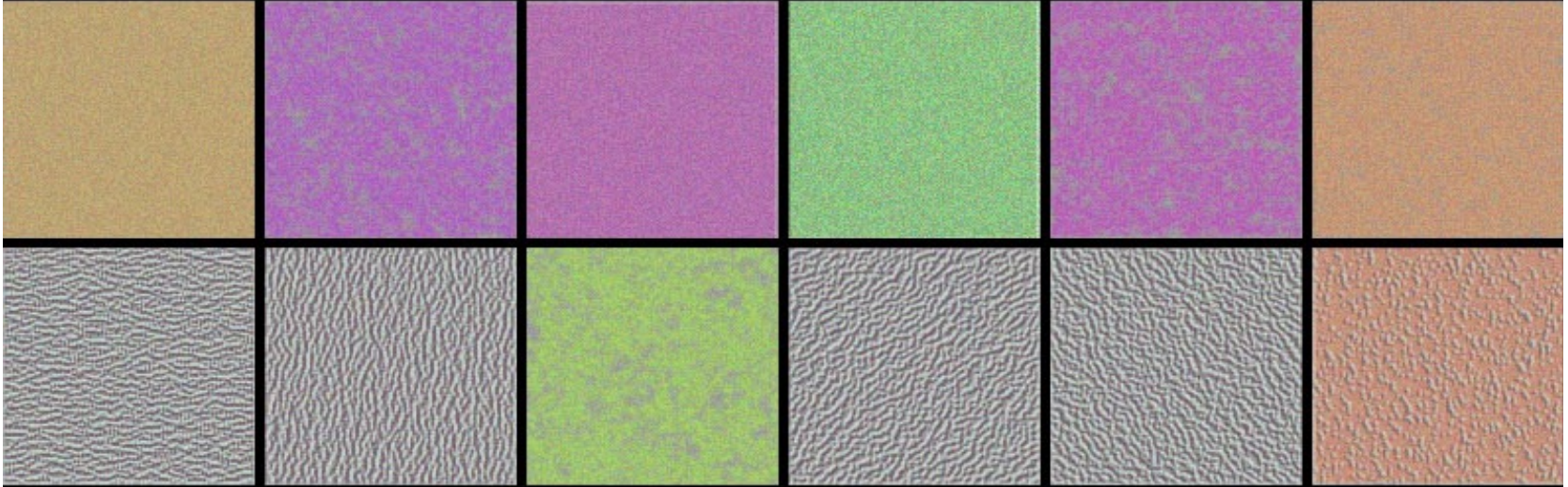


Architecture of VGG-16

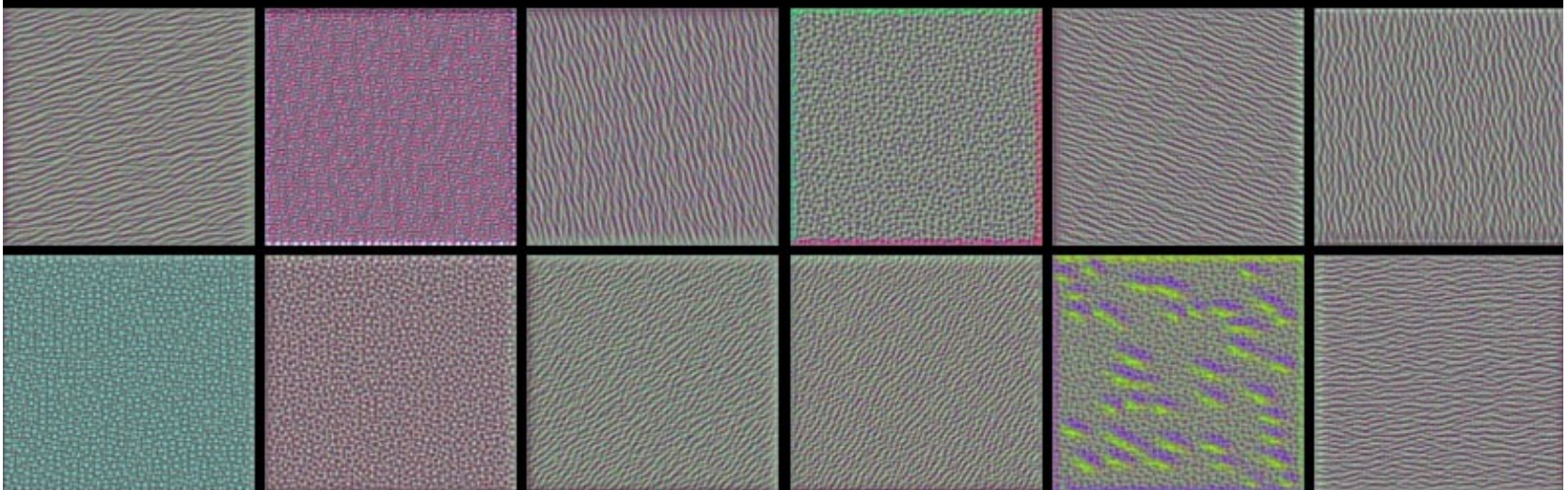


Visualize VGG-16 filters: inputs which maximize each filter

conv1_1: a few of the 64 filters

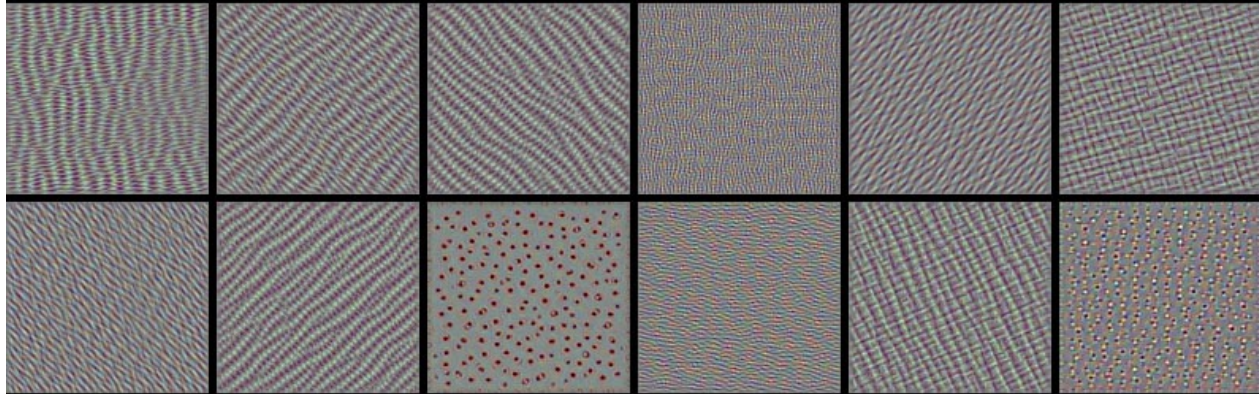


conv2_1: a few of the 128 filters

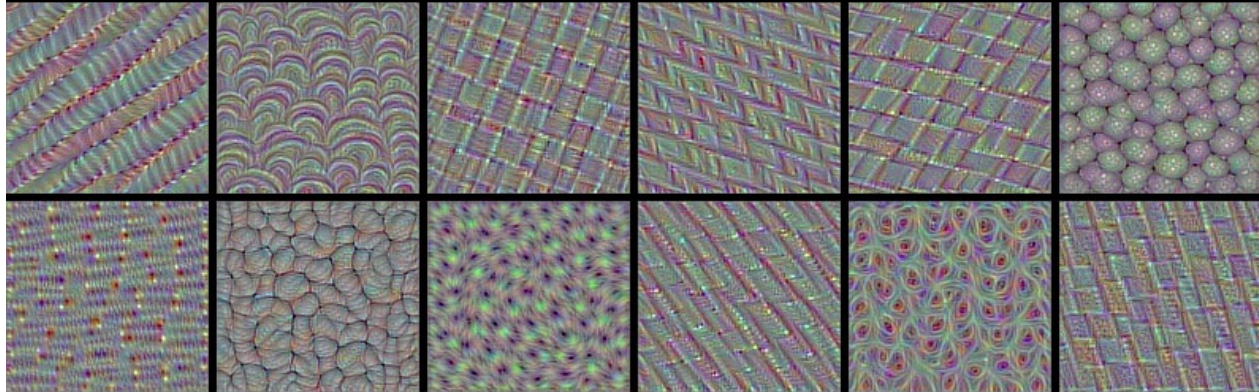


- Many filters are identical, but rotated.
- High-level filters incorporate more complicated patterns from a larger receptive field
- Highest layers capture textures of the objects for classifications such as feathers, eyes, etc.

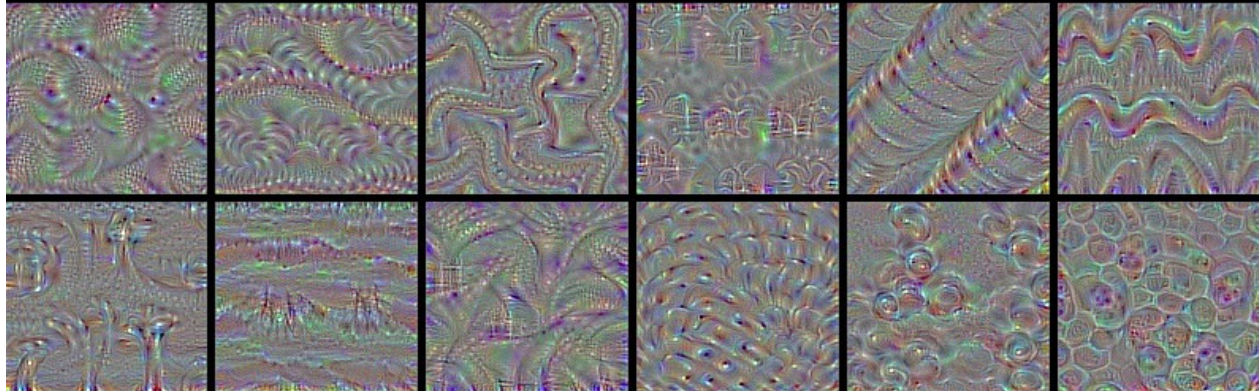
conv3_1: a few of the 256 filters



conv4_1: a few of the 512 filters



conv5_1: a few of the 512 filters



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GoogLeNet (2014)

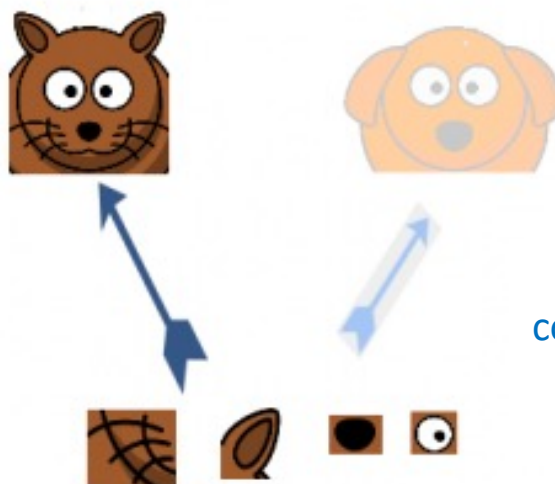
C. Szegedy, et al, Going deeper with convolutions, 2015

- ILSVRC'14 winner with 6.7% top-5 error
- Deeper network (22 layers vs 7 in AlexNet)
- Computational Efficient with less parameters (4M vs 60M in AlexNet)
- No FC layers

Main idea - Inception:

network decides best filter size

The layer should have the appropriate filter sizes to detect different objects.



Pooling
convolution
Softmax
others



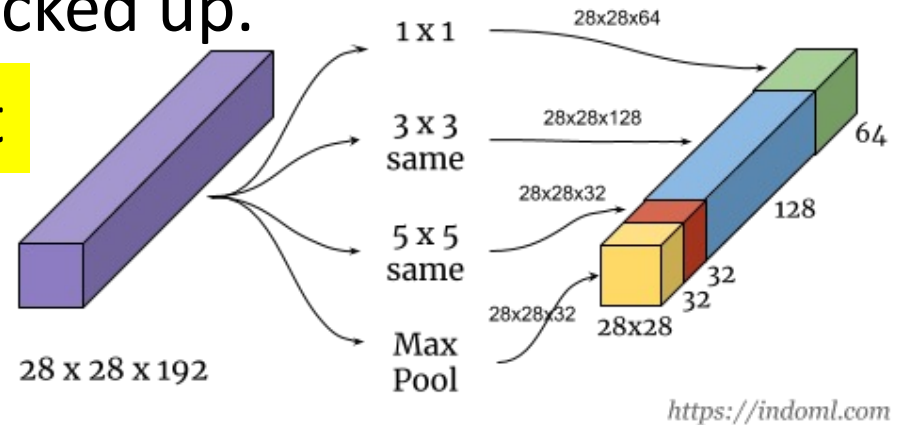
Inception

- Multiple filter sizes per convolution and the best filter size will be picked up.

Problem? **computation cost**

Cost for the 5x5 filter?

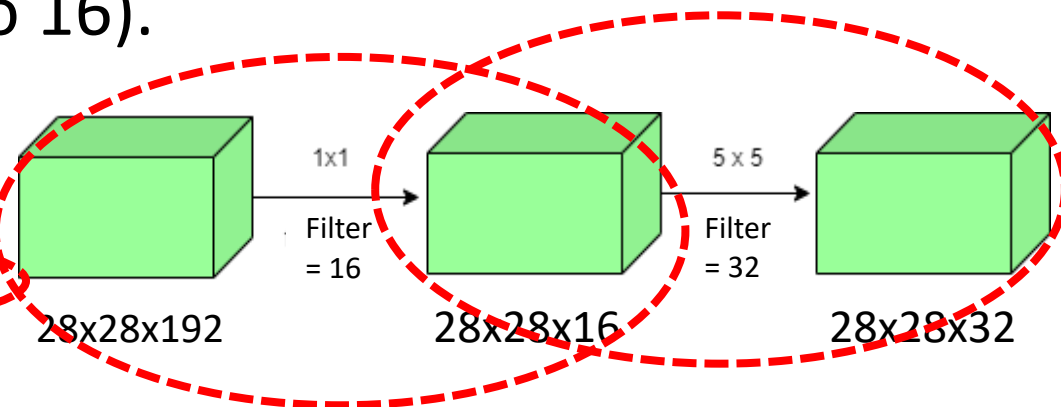
$$(5 \times 5 \times 192) \times (28 \times 28 \times 32) \\ = \sim 120 \text{ millions operations}$$



With 1x1 convolution to decrease the number of parameters (from 192 to 16).

Number of operations:

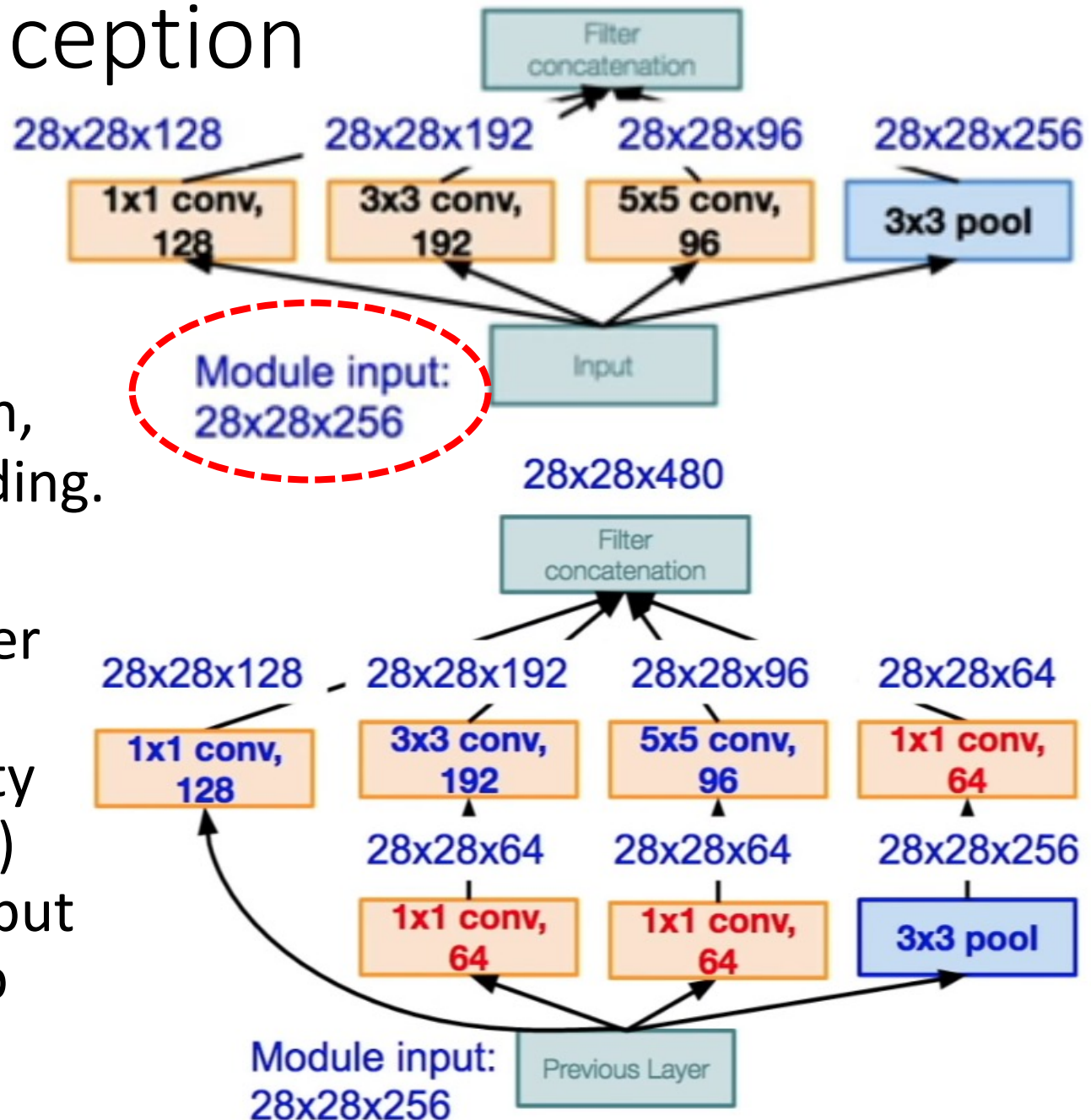
$$(1 \times 1 \times 192) \times (28 \times 28 \times 16) \\ + (5 \times 5 \times 16) \times (28 \times 28 \times 32) \\ = 12.5 \text{ millions}$$



Efficient Inception Model

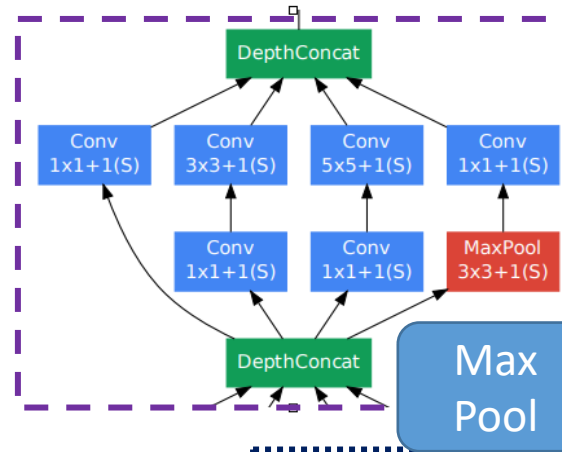
Maintain same image size
(height and width, 28x28) with padding.

Use conv 1x1 layer
(inception) to reduce complexity
i.e., (64 1x1 conv)
on 28x28x256 input to 28x28x64 map



GoogLeNet

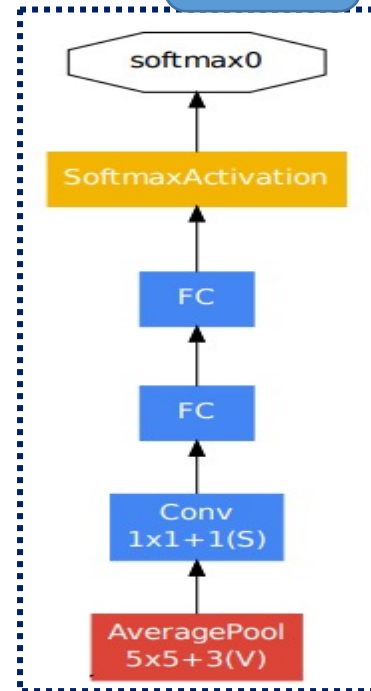
Inception modules sometime interleaved by Max Pooling layer



For good backpropagation:
some classifiers are injected
in the middle of the network.

Summary:

- deeper and efficient network
- 22 layers, 12x less parameters than AlexNet



Convolutional Neural Networks (CNNs)

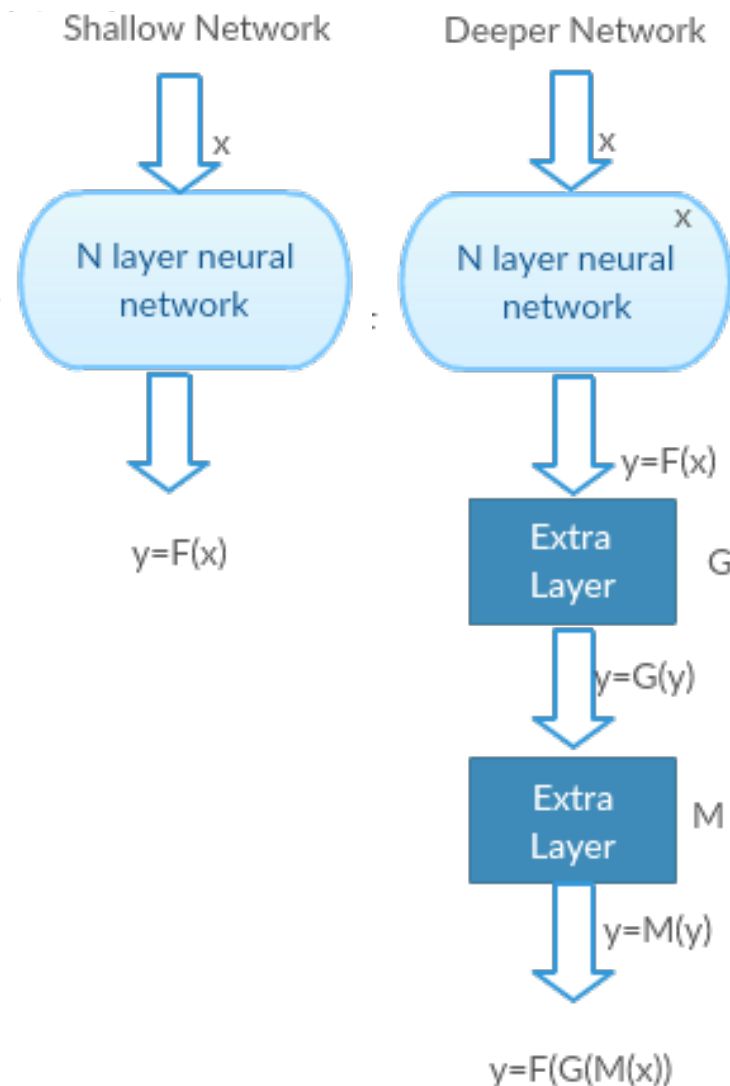
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Deeper Networks Perform Better?

- Theoretically, additional layers in deeper network can better approximate the mapping than its shallower counterpart
- At worst, the extra layers act as identity function
- In practice, *experiments reveal that deeper models doesn't perform well.*
- Using deeper networks is **degrading** the performance of the model.

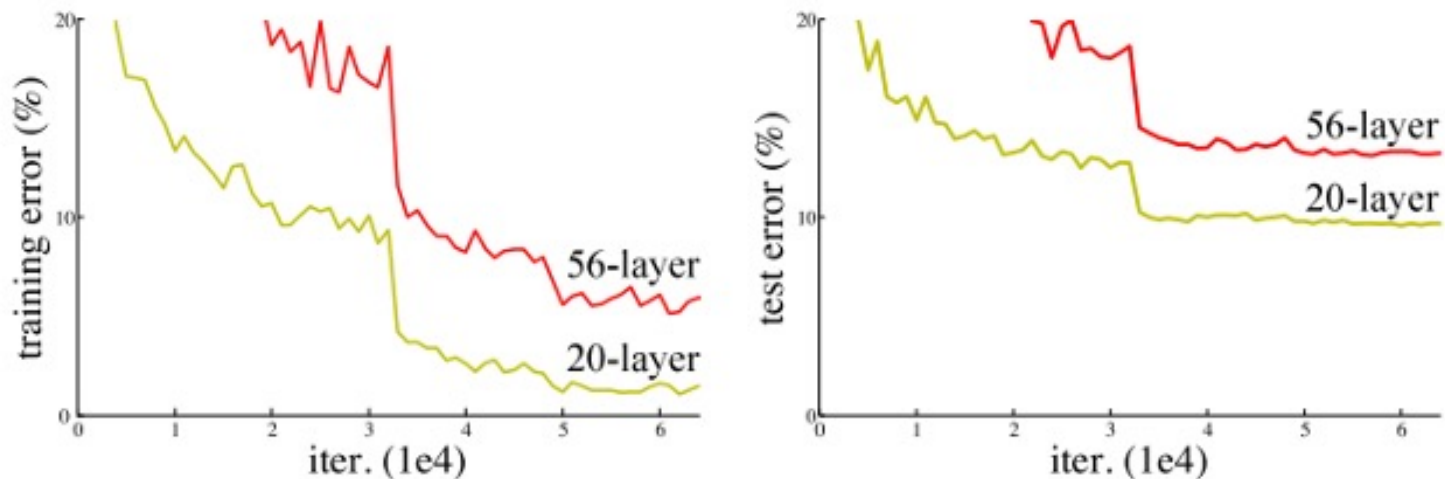


ResNet 2015

He, Zhang, Ren, Sun, Deep residual learning for image recognition.2016 (Microsoft).

Observations:

- A massive single mid-layer is sufficient to represent any function (last lecture).
- However, there is a common trend to have a deeper network architecture.
- Deep networks are hard to train because of the vanishing gradient problem



- adds an auxiliary loss in middle layers (GoogLeNet)

Residual Block

Stacking layers shouldn't degrade performance, as we could stack identity functions (no action)

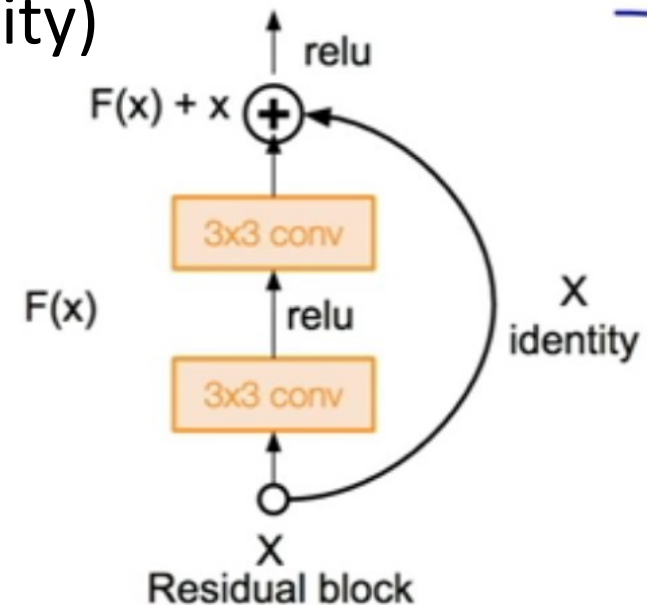
Hypothesis: **residual function $F(x) = H(x) - x$** is easier to fit than the desired underlying function $H(x)$

Easier to learn $F(x)=0$ than identity function

Use Skip/Residual Connection (identity)

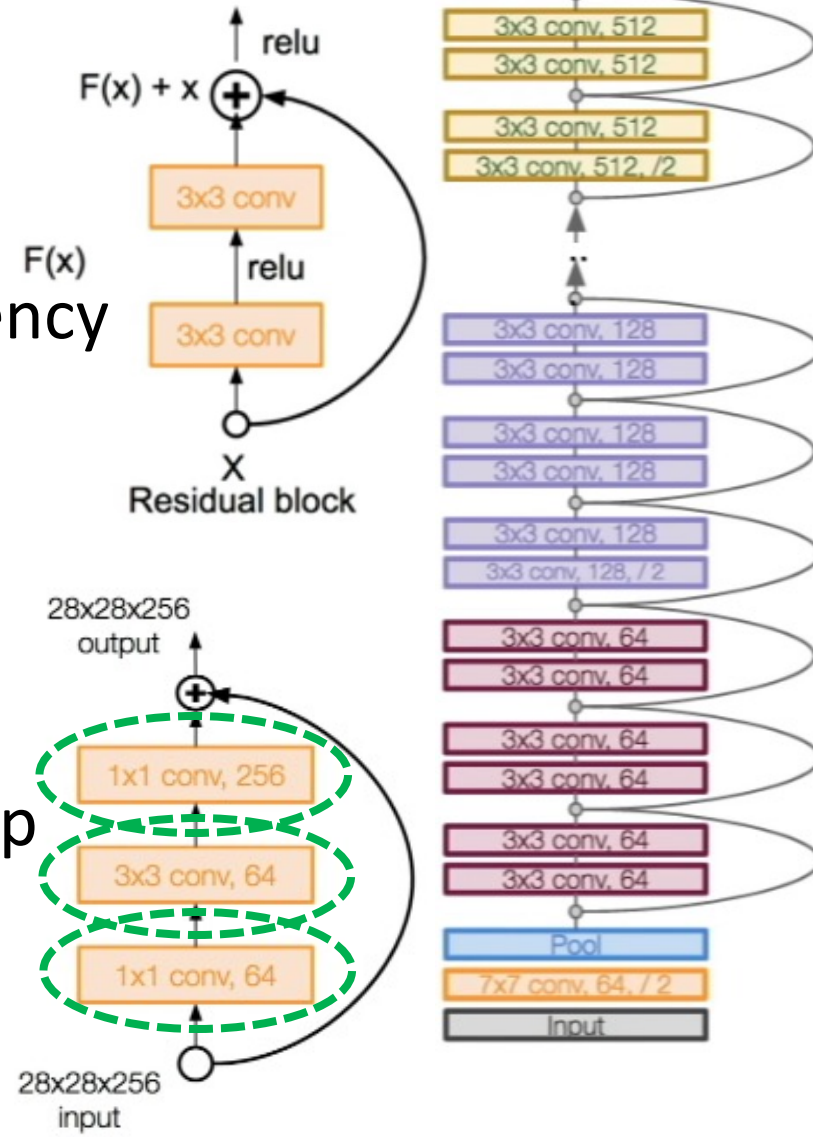
Advantages:

- Performance doesn't degrade with very deep network
- Ability to train very deep network
- Cheaper to compute



ResNet

- Stack “residual” blocks on top of each other
- Every residual block has two 3x3 conv. layers
- Periodically double # of filters and then pooling with stride 2
- Similar to GoogLeNet for efficiency
 - use 64 1x1 conv layer project input to 28x28x64.
 - the 3x3 conv is less expensive (depth 64 < 256).
 - Then use 256 1x1 conv to get a 28x28x256 feature map.
- ResNet is able to train very deep networks without degrading: 152 layers layers on ImageNet.



Convolutional Neural Networks (CNNs)

- What is Computer Vision?
- Milestones in Computer Vision
- What is Convolutional Neural Networks (CNNs)?
 - Convolutional layer – Filter
 - Pooling
 - Dense layer
- Image Classification
- AlexNet 2012 and ZFNet 2013
- VGGNet 2014
- GoogLeNet 2014
- ResNet 2015
- **ReduNet 2022**

References

- MIT online course 6.S191 “Introduction to Deep Learning”
- Stanford CS230 “Deep Learning”
- Chan, Yu, You, Qi, Wright, Ma, ReduNet: A White-box Deep Network from the Principle of Maximizing Rate Reduction, Journal of Machine Learning Research 23 (2022) 1-103

ReduNet – White-Box Approach

Chan, Yu, You, Qi, Wright, Ma. ReduNet: A White-box Deep Network from the Principle of Maximizing Rate Reduction. Journal of Machine Learning Research 23 (2022) 1-103

Conventional Deep NN is a “black box”

i.e., can't be explained
NN approximates an
unknown function $x \rightarrow y$
with a composition of
unknown functions

$$f(x) = g \circ f_L \circ \dots \circ f_2 \circ f_1(x)$$

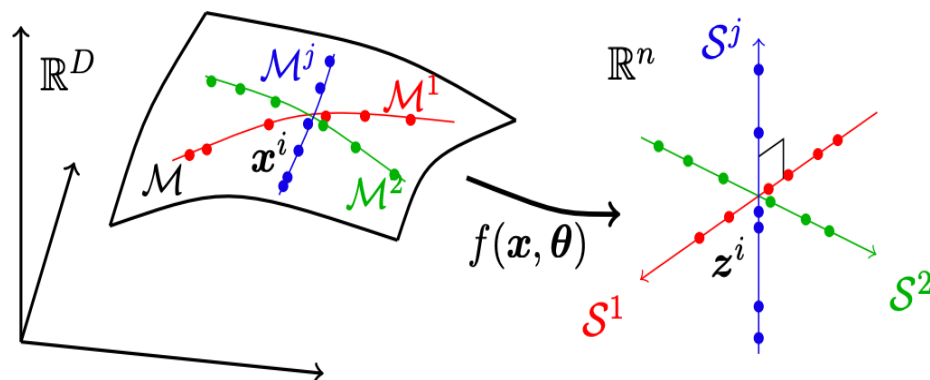
Desired properties:

- **Within-Class Compressible**: Features of same class are highly compressed in **low-dimensional** linear subspace.
- **Between-Class Discriminative**: Features of different classes are in highly **incoherent** linear subspaces.

ReduNet takes a “white box” approach

i.e., can be explained
ReduNet learns a particular
function $f(x, \theta): x \rightarrow z(\theta)$
to maximize

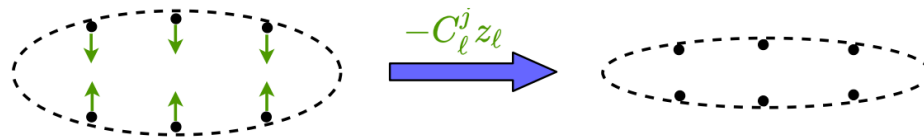
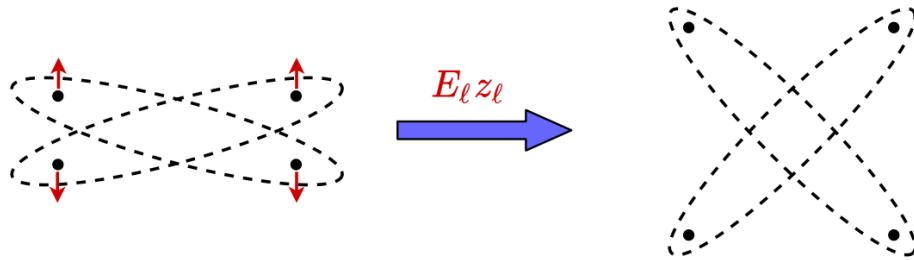
ΔR , Rate Reduction



ReduNet

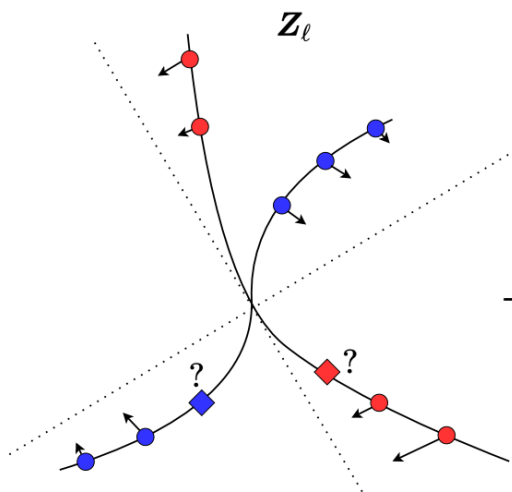
$$\max_{\theta} \Delta R(Z) = \max_{\theta} (R - R^c)$$

$$\text{Gradient } \partial \Delta R / \partial Z = \underbrace{E_l z_l}_{\text{expansion}} - \underbrace{\sum C_l^j z_l}_{\text{compression}}$$

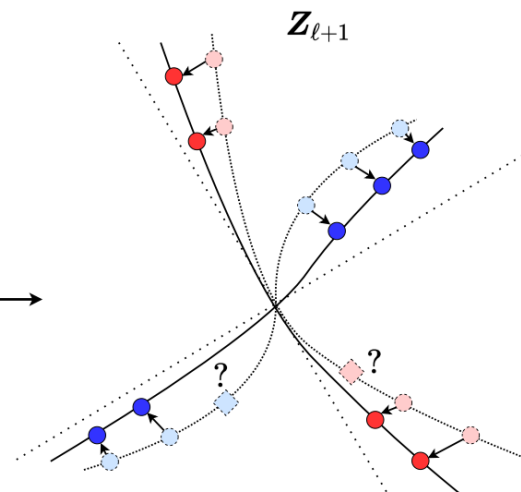


- push different classes apart

- flatten data of each class into a subspace



$$Z_{\ell+1} = Z_{\ell} + \eta \cdot \left. \frac{\partial \Delta R}{\partial Z} \right|_{Z_{\ell}}$$



Various Computer Vision Tasks

- **Visualizing and Understanding CNN**
- Object Localization
- Object Detection
 - Sliding windows
 - R-CNN, Fast R-CNN, Faster R-CNN, YOLO
- Semantic Segmentation – Mask R-CNN

Visualizing and Understanding CNN

Zeiler and Fergus, Visualizing and Understanding Convolutional Networks, 2013

Learn how CNN understand images

Specific Questions

1. Visualization: What patterns in the image will activate feature maps at different layers?
2. Can visualization help to design better convolutional networks?

3. Can NN identify the intended object?

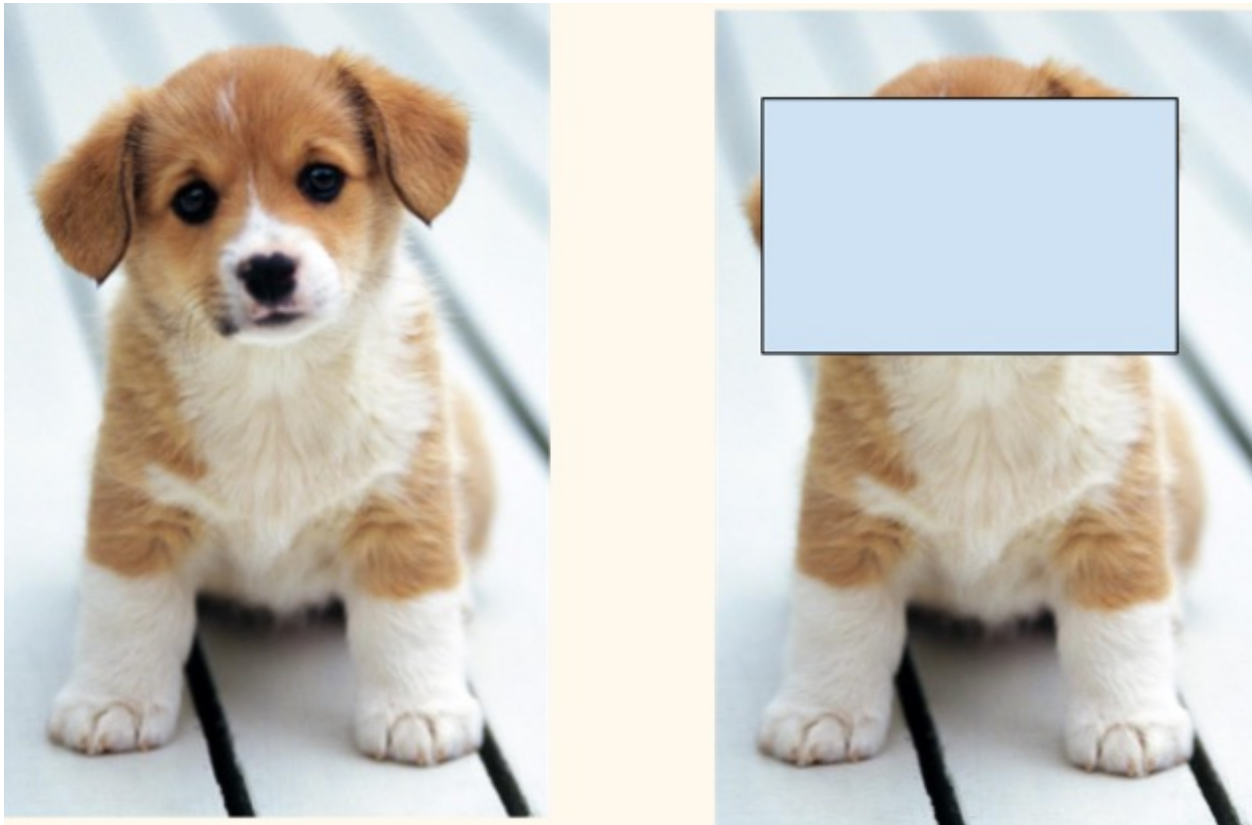
That is, recognizing the object by correct features, as opposed to predicting the object based on its surroundings.

4. Are CNNs computing relationships of features?

For example, how a nose is spaced relative to the mouth and eyes.

1. Occlusion Sensitivity

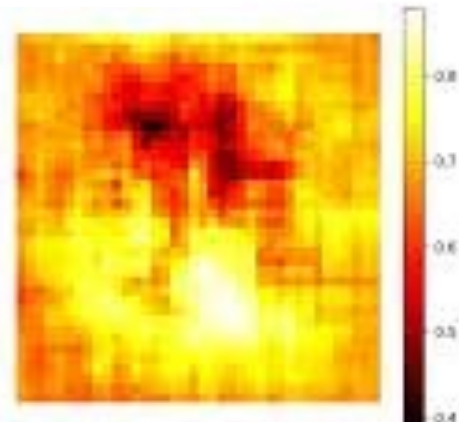
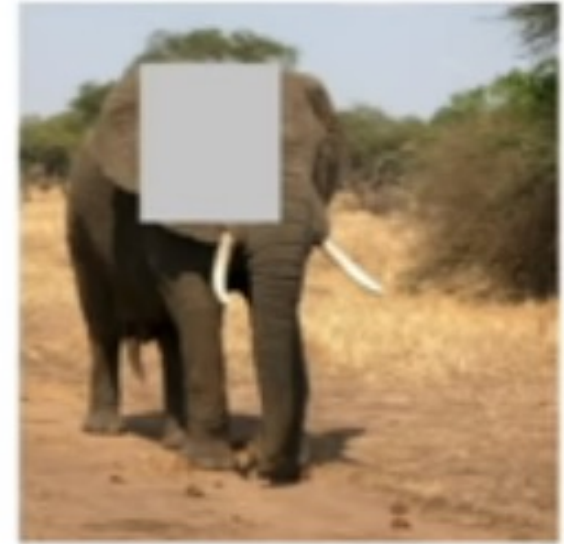
- Mask an object's important part in the image.
- If prediction accuracy doesn't suffer (i.e., NN still predicts dog with the same accuracy), then NN isn't learning specific features of the object.



Probability Heat Maps/ Saliency Maps

Which part of the image are most important for prediction?

- Mask part of the input image (with a grey patch for example). Repeat multiple times by sliding the patch over the image.
- For each pixel, compute probability of predicting the class correctly when the pixel is masked.
- **Probability heatmap** - the darker the region the lower the probability of correct classification, the more important is the pixel.



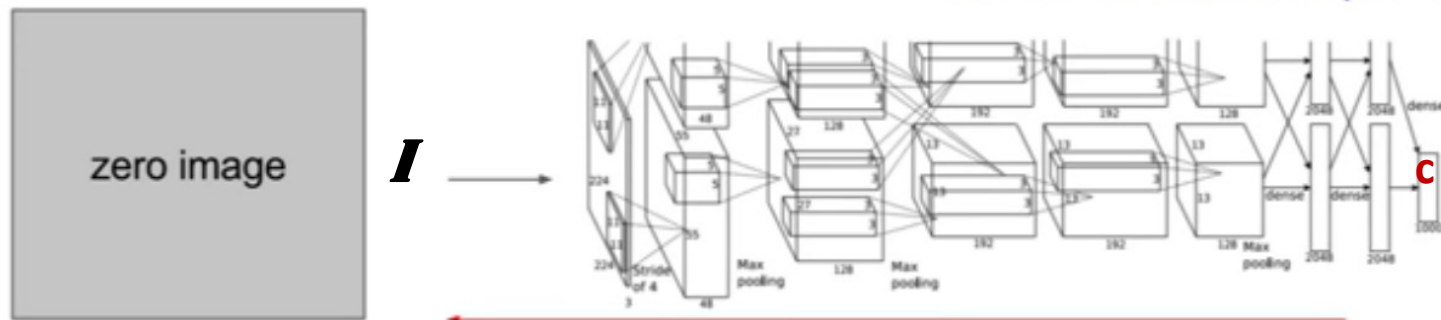
2. Gradient Ascent

Fix all weights of a CNN. Generate a synthetic image that maximizes the activation value of a neuron.

Initialize image to zeros

$$\arg \max_I \boxed{S_c(I)} - \lambda \|I\|_2^2$$

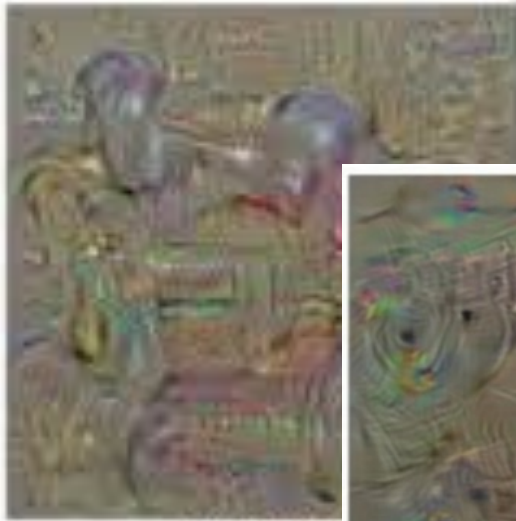
score for class c (before Softmax)



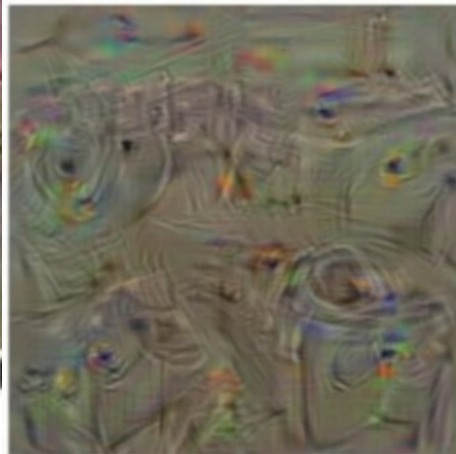
Repeat:

1. Forward image to compute current scores
2. Backprop gradient of neuron value w.r.t. image pixels
3. Make a small update to the image

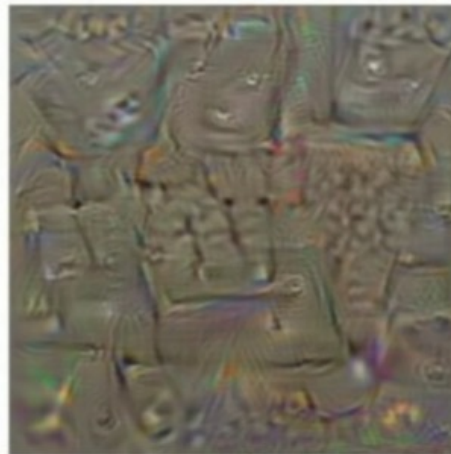
Generated Images with Gradient Ascent



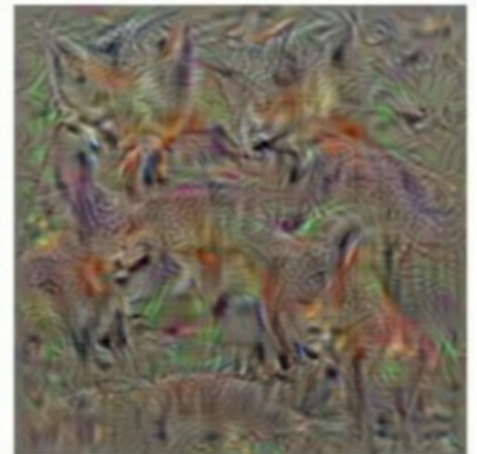
dumbbell



washing machine



computer keyboard



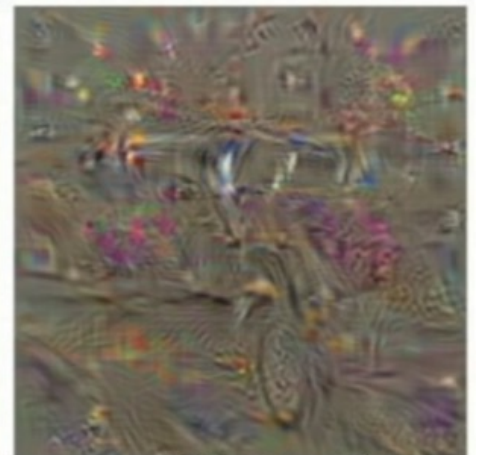
kit fox



goose



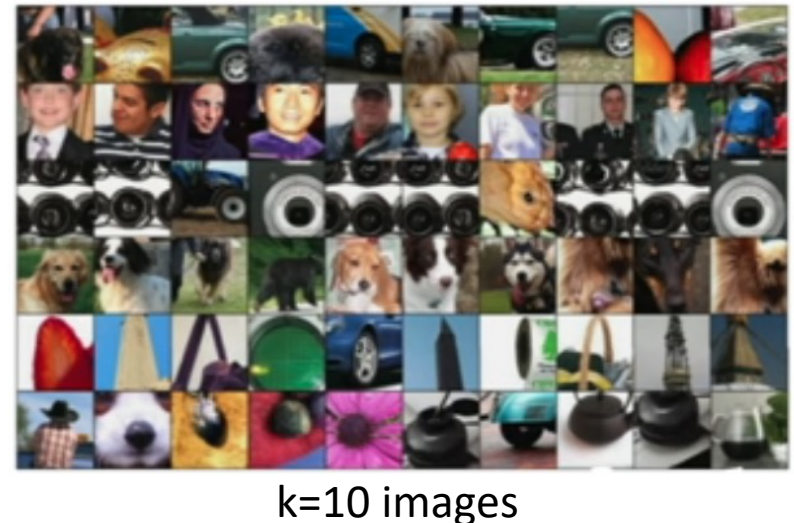
ostrich



mouse

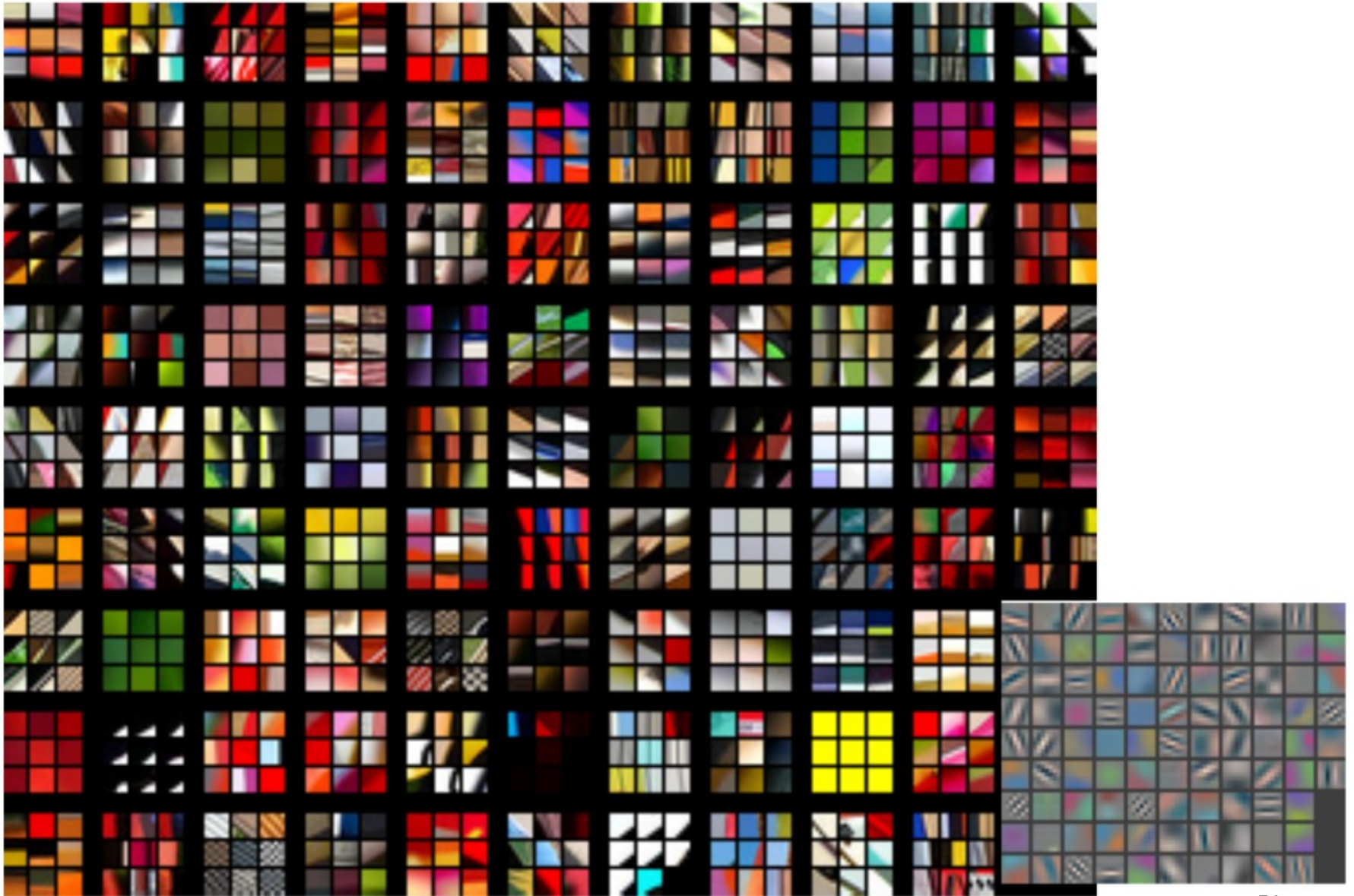
3. Maximally Activating Patches

- Pick a neuron, run many images through the network and record activation values for that neuron.
- Consider the top k images that results in largest activation values for the neuron.
- Show top k images (patches or LRF) of that neuron.



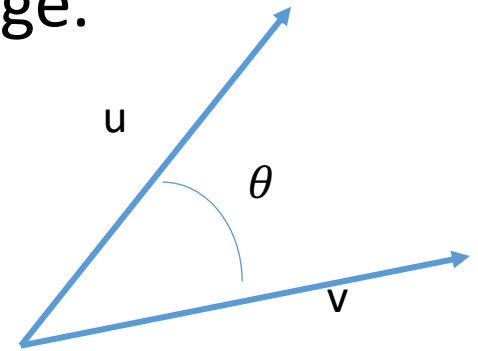
- Each row corresponds to a neuron.
Higher-level neurons (right) have larger LRFs

Layer 1 – top-9 patches (96 neurons/filters)

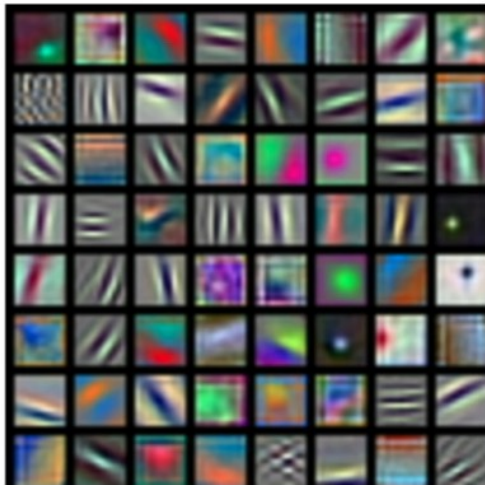


First layer Convolution Filters

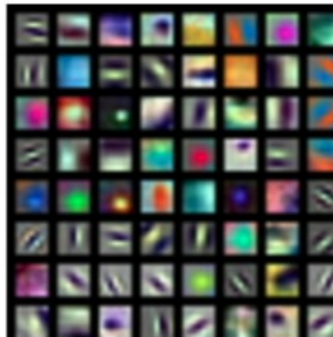
- Convolution filter computes “dot” product between the filter and its LRF in the image.
- Dot product: $u \circ v = |u| |v| \cos(\theta)$
 - measures similarity of u and v
 - maximum when $\theta = 0$, i.e., when $u = v$.
- Maximum when LRF matches filter.



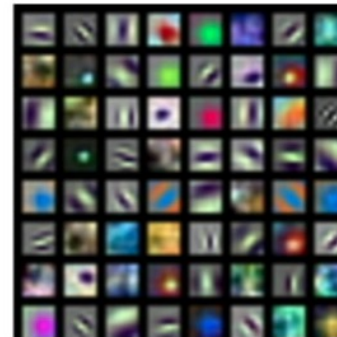
$$w^T x = \sum_{ij} x_{ij} w_{ij}$$



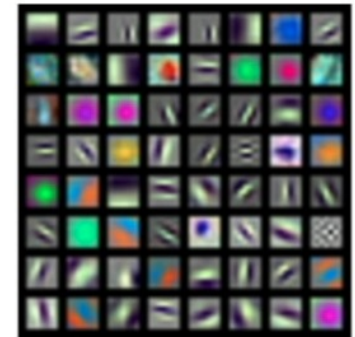
AlexNet:
64 x 3 x 11 x 11



ResNet-18:
64 x 3 x 7 x 7

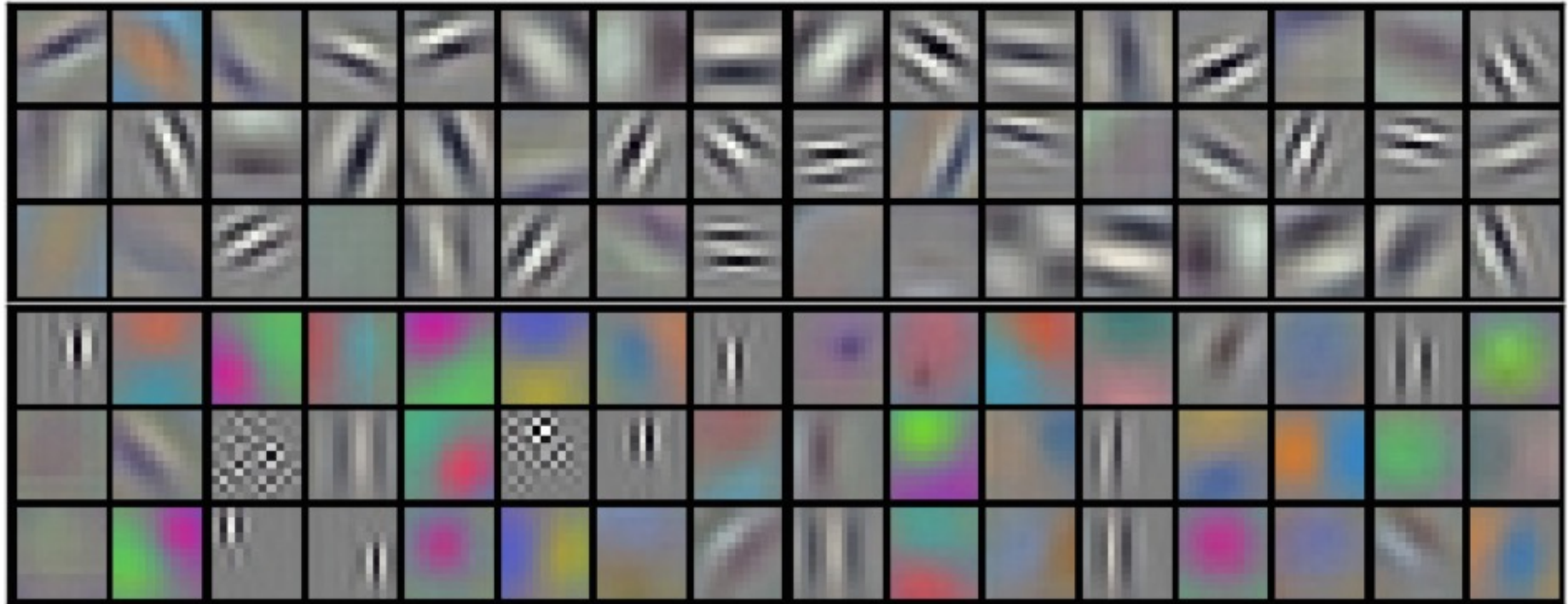


ResNet-101:
64 x 3 x 7 x 7



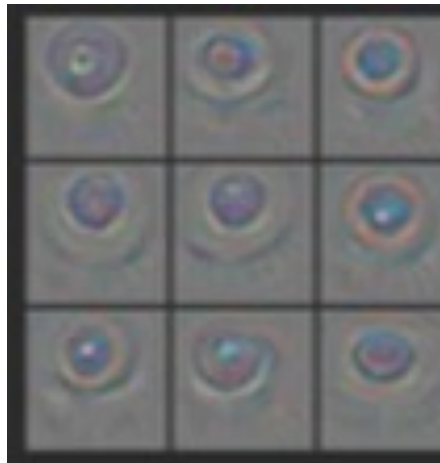
DenseNet-121:
64 x 3 x 7 x 7

Layer 1 – templates for edge detection (96 filters: size 11x11x3)



Layer 2

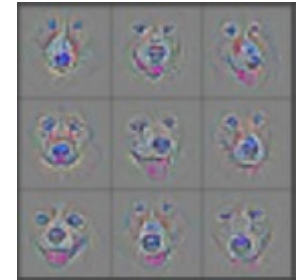
- Take a random feature map
- Find the top-9 patches from the validation images that give maximal activation of the feature map
- Then reconstruct the patterns in the patches that caused high activations in the feature map.
- Why different?
fainter because of ReLu and pooling



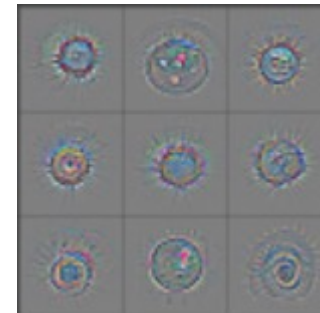
Layer 3



Layer 4



Layer 5



How to find similar pictures?

Similarity Measures (Euclidean distance)

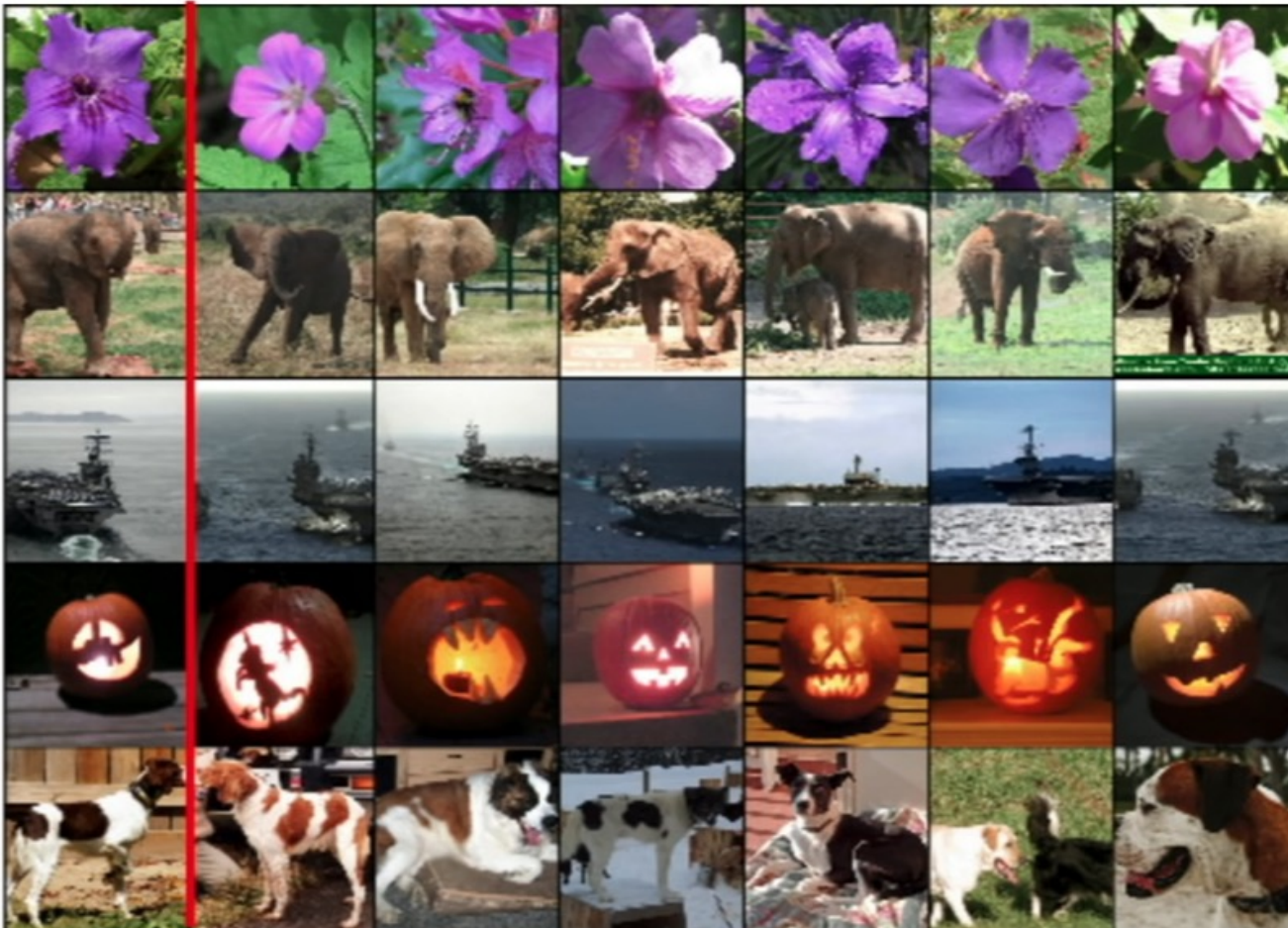
Find the nearest neighbors in pixel space,
i.e., sum of square differences of each pixel



Pictures
look similar
but are
different
objects.

Last layer (feature vector)

Compute **Nearest Neighbors** of the feature vectors and retrieve their images



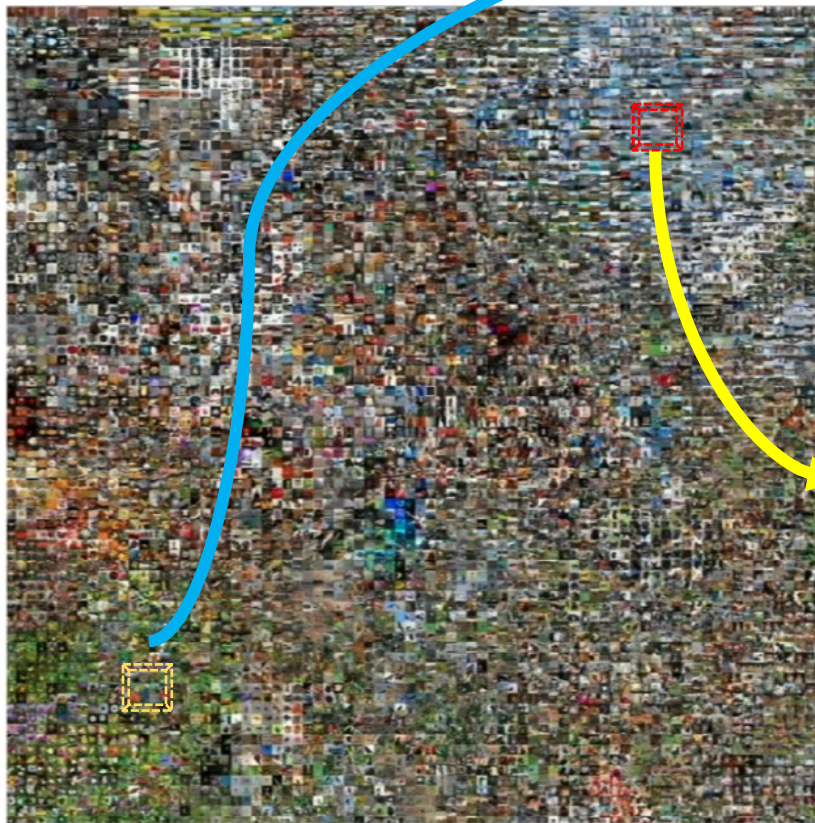
Images look different, but are similar in feature space.

For example:
Dog images
(similar in feature vectors)
are different
in positions,
crops, colors, etc.

ILSVRC 2012 Data

<https://cs.stanford.edu/people/karpathy/cnnembed/>

50,000 validation images reduce
with 4096-dim feature vectors
to 2-dim

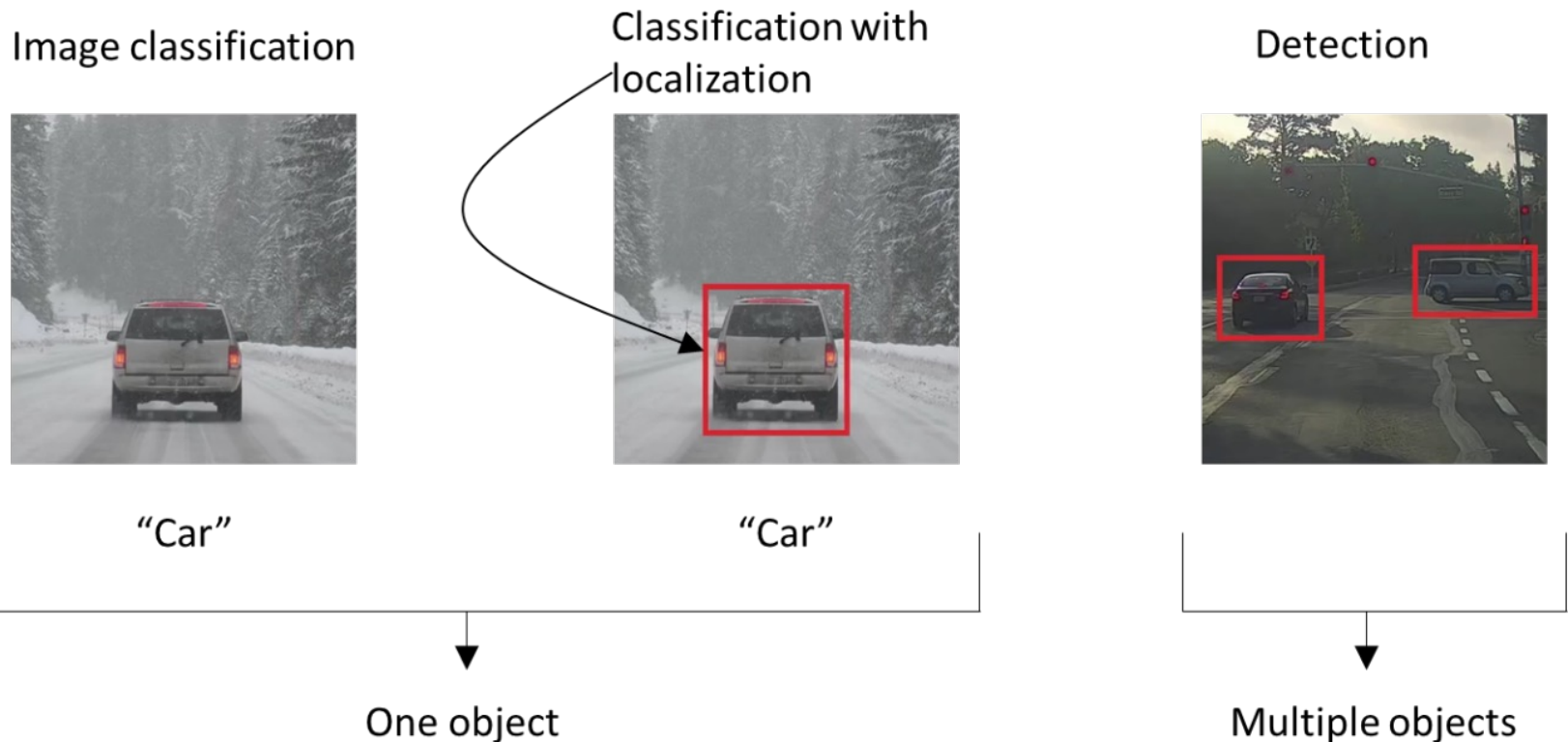


Various Computer Vision Tasks

- Visualizing and Understanding CNN
- **Object Localization**
- Object Detection
 - Sliding windows
 - R-CNN, Fast R-CNN, Faster R-CNN, YOLO
- Semantic Segmentation – Mask R-CNN

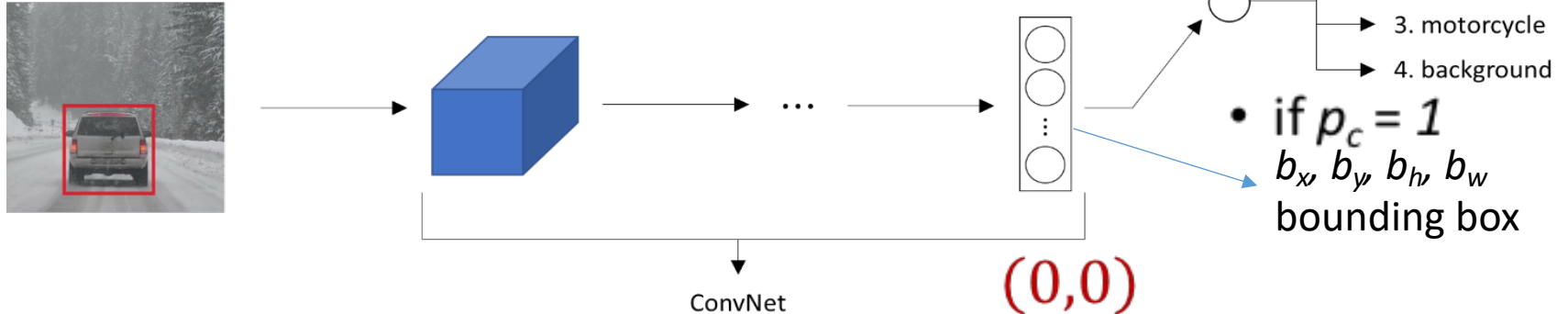
Object Localization

- **Image classification** – identify an object in a picture
- **Classification with localization**
 - identify an object with a bounding box
- **Detection** - detect and localize all objects (usually >1)
e.g., cars, pedestrians, bicycles and motorcycles



CNN Localization and Classification

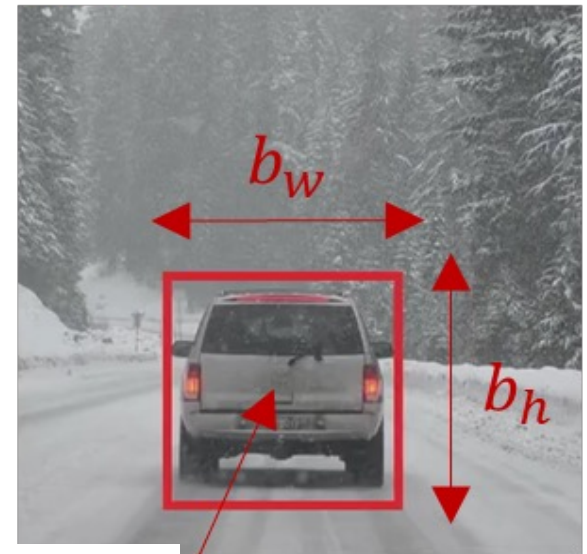
Assume a NN for classifying objects
(pedestrian, car, motorcycle, background)



Four numbers, b_x, b_y, b_h, b_w specify the bounding box.

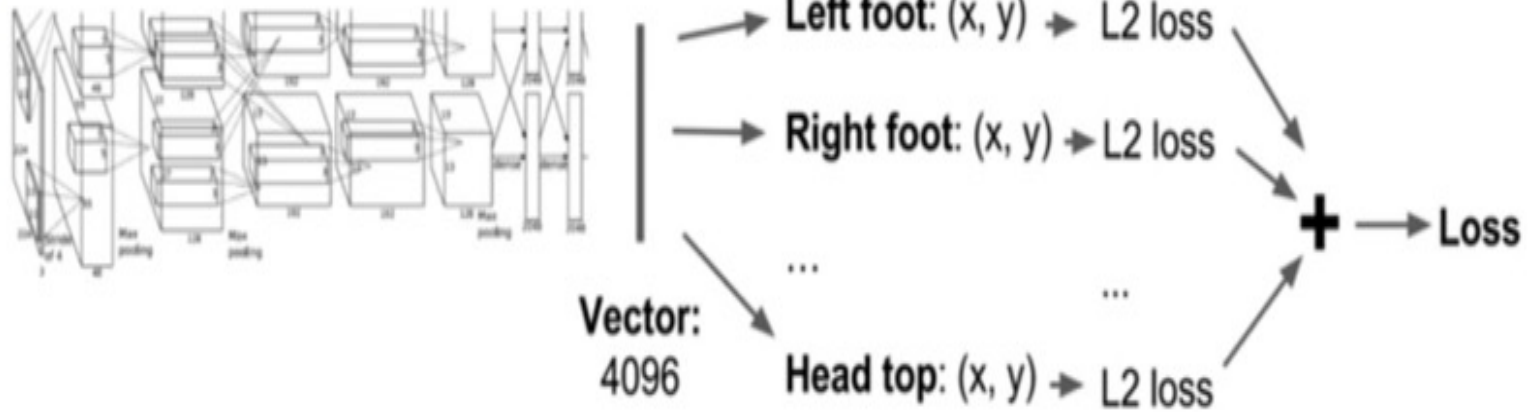
Example: $b_x = 0.5$; $b_y = 0.7$; $b_h = 0.3$; $b_w = 0.4$

- Output: $(p_c, b_x, b_y, b_h, b_w, c_1, c_2, c_3)$
where *if no object (background)*, $p_c = 0$
if $p_c = 1$, then one of $c_1, c_2, c_3 = 1$
- Training set needs bounding boxes information and labels.
- Loss: $\lambda L_1 + L_2$ (L_1 =classification error; L_2 = bounding box error)



b_x, b_y $(1,1)$

Pose Detection



$$\begin{array}{c}
 l_{1x}, l_{1y}, \\
 \vdots \\
 l_{16x}, l_{16y}
 \end{array}$$

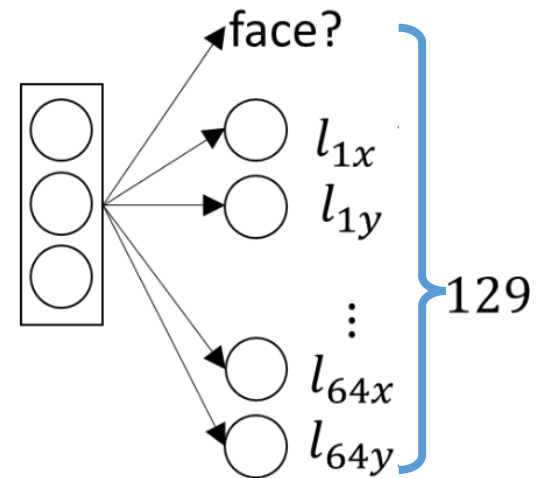
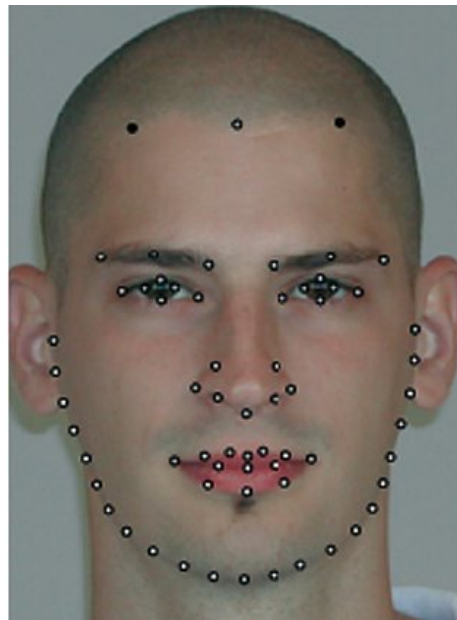
Landmark Detection

- Landmarks are coordinates of important points, e.g., Face recognition, positions of eyes



$$\left. \begin{array}{l} l_{1x}, l_{1y}, \\ l_{2x}, l_{2y}, \\ l_{3x}, l_{3y}, \\ l_{4x}, l_{4y}, \\ \vdots \\ l_{64x}, l_{64y} \end{array} \right\} x, y$$

→ ConvNet →

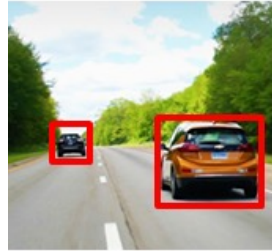


Various Computer Vision Tasks

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Object Detection

First create a labeled training set with cropped images



training set:



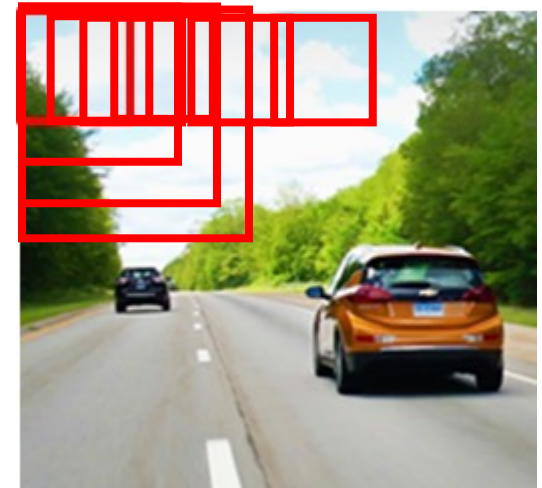
→ ConvNet → y

Sliding window detection

- Start with a little square at the top-right corner and check whether the little square contains a car
- Repeat by sliding the red square across the whole image
- Repeat with different sizes of square

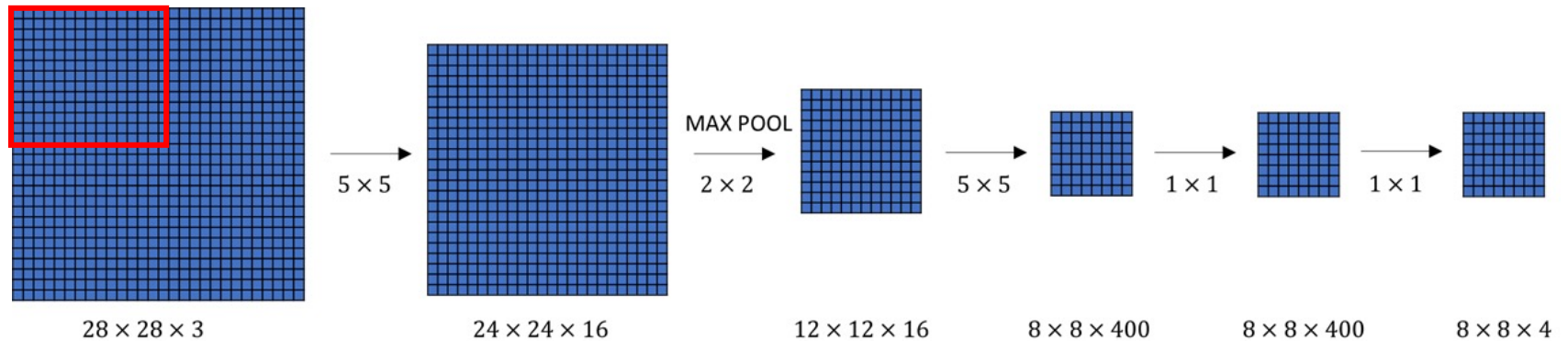
Disadvantage: huge computational cost

Too big stride or too big step size reduce computational cost, but hurt performance, i.e., miss the car.



Convolutional Operation of Sliding Windows

Convolutional implementation on a 28×28 image



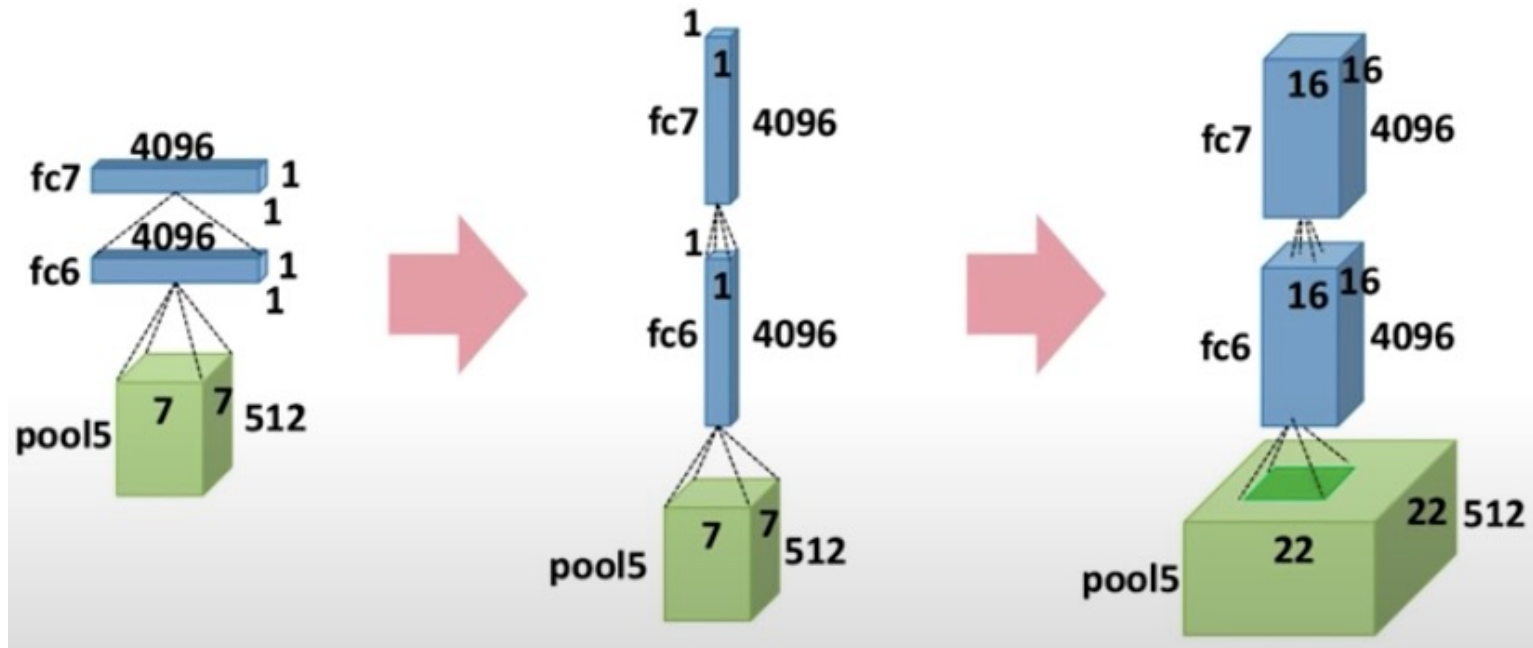
Equivalent to sliding 14×14 windows (stride 2) on a 28×28 image.



End up with an $8 \times 8 \times 4$ output

Convolutional NN for FCN

- Another pictorial view about the process



Fully connected layers

Convolution layers

For the larger Input field